Department of Electronic Engineering,
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# Cybercrime Investigations of AI-Enabled Environments: “When AI Turns Rogue: Forensic Challenges and Innovations in AI-Powered Cybercrime

Candidate Number: 2507799

A Dissertation Submitted
In Partial Fulfilment of the
Requirements for the Degree

**MSc in Project Management**

Supervisor: Richard Granger

06/09/2025

## Author’s Declaration

I am proud to declare that this dissertation is the result of my own original research and writing.

I confirm that this is the final and complete version of my work, incorporating all necessary revisions as approved and accepted by my supervisor(s).

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## Abstract

*We can only see a short distance ahead, but we can see plenty there that needs to be done." —* **Alan Turing, the father of modern computer science**. Artificial intelligence (AI) has evolved from an emerging technology to become a fundamental tool which cybercriminals use to obtain automated, programmable and adaptable attack tools. The development of AI technologies, which offer transformative benefits in proper contexts, has led to their misuse by malicious actors for executing sophisticated cyberattacks. The research examines the forensic effects of AI-based cybercrime through critical analysis while presenting new methods to improve investigative work and forensic evidence handling, and AI crime attribution processes.

A mixed-method research design has been undertaken, combining both secondary and primary data to give depth and breadth of analysis. We searched and found 165 academic, industrial, and government papers on AI-based cybercrime and response. The former entailed five semi-structured interviews with leaders in the field of cybersecurity and forensic research and was augmented with non-participatory observation of dark web-based AI tools and platforms. This three-way pulling strategy provided both a big picture approach to theory development and a small picture approach with practitioner savvy. The findings reveal that AI has already established itself as a compact element of the current cybercrime. It has also been abused through speech phishing through natural language models, deep fake-based income frauds, self-updating malware that mutates behaviour properties in real-time, and adversarial examples swindling detection models. The implications of these capabilities pose multidimensional forensic challenges such as black-box obscurity, the volatility of digital evidence, attribution, and ambiguities related to the admissibility of AI-created evidence (in court). Dark web research identified a booming black market where AI academic breakthroughs are already used to commit crimes, in some cases within a few weeks. This dissertation proposes a four-facet forensic adaptation framework, to overcome these challenges, that includes: (i) Explainable AI (XAI) integration to favour transparency and evidentiary alignment, (ii) blockchain-enhanced chains of custody to preserve artefact integrity, (iii) cross-sector/transnational intelligence exchange to favour preparedness, and (iv) forensics-specific AI regulation rules to direct investigators, corporations and policymakers. It has been concluded that the adaptation of forensic science to suit AI-enabled crime is a necessity and no longer an option

## Acknowledgments

Completing this dissertation was one of the most challenging and rewarding experiences of my life, and it would not have been possible without the support of some truly wonderful people. First and foremost, I want to thank my supervisor, Richard. Your door was always open, and your insights consistently pointed me in the right direction. Thank you for your patience and for believing in this project from the very beginning.

My research was immeasurably enriched by the cybersecurity experts who shared their valuable time and knowledge with me—thank you for your crucial contributions. To my supervisor, family and friends, I don't have the words to thank you enough. For putting up with my late nights and endless talk about AI, and for being my biggest cheerleader, I am eternally grateful to my supervisor. This is for them.

# Research Ethics Approval

DRec approval.pdf

**EE – DEPARTMENTAL RESEARCH ETHICS COMMITTEE**
**REVIEW OUTCOME**

| | |
|---|---|
| Student Name: | Dhanush Gowda Gowdahalli Prasad |
| Student ID: | |
| Course Title/Year/Cohort (Sep or Jan) | Sep 2024 |
| **Title:** | |
| Cybercrime Investigations of AI-Enabled Environments: When AI Turns Rogue – Forensic Challen[g]es in AI-Powered Cybercrime | |
| **Summary of the Research Activity: (As documented by student in Research Ethics Form)** | |
| The research critically investigates how rogue AI is being exploited in cyberattacks and explores forensic challenges associated with such incidents. It also examines how digital forensic tools an[d] frameworks can be enhanced to detect, mitigate, and respond to threats posed by AI-generated cybercrime. The study employs a mixed-methods design with secondary data analysis, expert int[erviews] and observational research on the dark web. | |
| **Points Raised by Committee (If needed more clarity from student)** | |
| 1.<br>2.<br>3. | 4.<br>5.<br>6. |
| **Questions for the Student: (If conflict of interest exists and needed more clarity from student)** | |
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| **Risks involved in this Research: (According to EE DREC)** | |
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**For EE Departmental Research Ethics Committee Only**

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|---|---|
| Student Name and Number: | |
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| | Specified Condition (If Applicable): |
| Authorised by Chair EE DREC / Director of PGT Education | *Dr Syed M Hasan* – Chair EE DREC |

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## United Nations Sustainable Development Goals

The Sustainable Development Goals (SDGs) provide the world with a blueprint for achieving a better, safer, and more equitable future. The dissertation directly connects with two SDGs, as it discusses both the forensic issue of AI-enabled cybercrime and the governance issue of AI-enabled cybercrime, and proposes creative solutions to enhance infrastructure, justice, and institutional resiliency.

**SDG 9: Industry, Innovation and Infrastructure**

This program enables SDG 9 by creative thoughts about computer forensics and computer resilience. With AI going systemic (critical infrastructure, finance, healthcare, supply chains, and government systems) and embedded, its misuse has systemic risks. The paper contributes to the creation of future-resilient, hardier, and trusted infrastructures by shaping forensic thematic instruments such as explainable AI, blockchain chain-of-custody and intelligent-sharing apparatus. And these kinds of advances transform even such industries, which are non-digital, to tread in measures as delicate as a cosmetic without ever actually touching the digital trust of a person, and most of the technological progress takes a more dangerous turn than the one it is used to.

**SDG 16: Peace, Justice and Strong Institutions**

Undertaking justice-related projects such as this also helps to meet SDG 16 by responding to the justice gap that emerges due to opaque, volatile, or legally inadmissible AI-generated evidence. Existing judicial systems and forensic frameworks are incapable of attributing attacks perpetrated by AI and cannot presently authenticate AI-produced artefacts. Included in the solutions to improve accountability, transparency and evidential integrity suggested in the dissertation are explainable AI to interpret black-box systems and forensic AI evidence standards. Bolstering these capabilities will improve investigative, prosecution, and prevention of AI-enabled crimes by justice systems and international institutions to advance peace, the rule of law, and a robust governance structure

# Chapter 1 Introduction

## 1.1 Background Information.

Over the past decade, artificial intelligence (AI) has transitioned into the mainstream field that dominates the economy, security systems, and even criminal activities. What was once limited to research laboratories ended up on the open-source platform, being commercialised via cloud providers, and in the darker corners of the internet (Vaishnavi, 2025; Outpost24, 2025), where it can be sold. Although such developments have enabled innovation in a variety of industries and fields, including the fields of healthcare, finance, and logistical operations, cybercriminals have utilized such innovations to carry out attacks with hitherto unattainable precision, velocity, and volume (Brundage et al., 2018; Al-Azzawi et al., 2025).

The development of AI-based cybercrime has left ordinary defensive and investigative capacities behind. Attackers have started to use generative AI to perform even more hyper-personalised phishing campaigns, create deep fake materials to cheat, auto-propagate the most vulnerable searches, and even design adaptive jarring that can evade signature-based security (Mirsky et al., 2021; Marchal et al., 2024). In the black marketplaces of the Internet, malicious users share AI-enhanced impregnation packages, ransomware-as-a-service products, and auto-disinformation services-reducing the entry threshold into the complex infiltration (TRM Labs, 2025; Cyble, 2025).

Such trends bring multi-dimensional risks to both the government and the corporate realms. The effects of theft of intellectual property, disruption of supply chain, and disinformation directed by campaigns can cost companies with multinational corporations (MNCs), underlining the fact that even a mild breach of information might be enough to undermine confidence and leave a lasting scar on the reputation (NCSC, 2023; Kaspersky, 2024). Governance-wise, underlining the impossibility of identifying the attack when AI-generated evidence is present could allow hackers to evade prosecution and pollute the entire digital investigative work (Hall, 2022; Alam & Altiparmak, 2024).

The time-sensitive aspect of combating AI-enabled cybercrime is that this challenge compounds over time: the more these tools are out there, the faster they refine themselves using the data stolen by others, learning to better combat them, or by cooperating with other criminals (Iqbal et al., 2023; UK NCSC, 2024). Since world organisations and law enforcement agencies are rushing to play catch-up, new forensics solutions are needed to simply keep in step with rogue AI systems, even outwit them, ideally.

## 1.2 Significance and Motivation

AI-powered cybercrime isn't just a tech headache; it's a major threat that affects all of us. It has the power to damage our economy, shake people's trust in the systems we use daily, and even put our essential services at risk. This issue is a sensitive warning to the AI research community that it must deal with the dual use of its innovations. Autonomous vehicles, medical diagnostics, and natural language processing are also run by the same algorithms that can be weaponised to commit large-scale fraud, social engineering, or even sabotage infrastructure (Calo, 2017; Kshetri, 2025). Unless active precautions are taken, these research milestones can be appropriated into a blisteringly expanding repertoire of crime tools.

The consequences to multinational corporations (MNCs) are short-term and long-term. Better attacks undermine consumer confidence, cripple the work, and may cost the firms even billions of dollars (World Economic Forum, 2025; Beazley, 2023). The reputational impacts of an intrusion of AI-generated evidence, e.g. deepfake executive communications or falsified transaction logs, can be as disastrous as not only the confidence of the stakeholders but also investor relations (Roberts, 2018; NCSC, 2024). In a strategic sense, failure to identify, trace, and charge AI-related attacks leaves one with a constant weakness, which encourages more attacks by malicious actors.

The study is also in line with the United Nations Sustainable Development Goals (SDGs):

SDG 9: Industry, Innovation and Infrastructure SDG 9: Use industries, innovation, and infrastructure in sustainable innovation and safe infrastructure to facilitate trust in the adoption of AI.

SDG 16: Peace, Justice, and Strong Institutions Development of strong forensic abilities to uncover and prevent AI-enabled crime contributes to justice systems and decreases the lack of accountability and strengthens international governance systems.

This study gives solutions to a vital gap in knowledge because of incorporating the three different angles of technical, managerial, and policy-oriented perspectives. It not only seeks to promote the field of forensic science into the age of AI but also offers some practical recommendations by design to those individuals developing, implementing and controlling the applications of AI technologies in critical fields.

## 1.3 Aims and Objectives

This dissertation is an attempt to objectively examine the forensic challenges that AI-based cybercrime presents and suggest novel methods of investigation capable of addressing them in realistic situations of high stakes. This objective takes into consideration both the technical requirements of mitigating rogue artificial intelligence systems, as well as the organisational need to minimise their effect on multinational operations and digital trust.

Research Questions (RQs-).

1. Which kinds of artificial intelligence-enabled cyberattacks are going on?
2. What are the major forensic, technical and managerial difficulties in investigating the AI-fueled cybercrime?
3. What are some of the ways of adapting digital forensic tools and frameworks, and how can they be properly managed to deal with the AI-motivated threats?
4. How effectively can Explainable AI (XAI) enhance transparency and increase the believability of forensic evidence?
5. What are some legal or ethical questions raised in examining autonomous or semi-autonomous AI systems?

**Research (ROs) focuses.**

RO1: Compare generative adversarial networks, deepfakes and automated intrusion systems to the other emerging risks of AI applications in cybercrimes.

RO2: Enumerate and critically present the technical, operational, and legal challenges that are currently encountered in digital forensics with reference to the use of AI.

RO3: assess whether an explainable AI (XAI) is present when it comes to enhancing transparency in investigation and credibility of evidence.

RO4: Formulate guidelines on implementing an effective framework based on current forensic frameworks to detect, analyse, and attribute AI-driven cyberattacks.

RO5: Explore ethical and legal connotations of gathering evidence, processing, and presenting it with the help of AI in the judicial frameworks.

These RQs and ROs were systematised clearly with the background and importance of the problem, and every objective leads to the general research question by its direct input.

## 1.4 Methodology

The study is mixed-method, where the secondary data and primary data will both be used to provide an in-depth review of the topic under investigation: AI-enabled cybercrime. The secondary research will be composed of a thorough review of peer-reviewed journals, industry reports, government publications and approved sources of threat intelligence, which will comprise a conceptual and empirical foundation to understanding emerging vectors of attacks, forensics problems and technological solutions.

The main research element consists of two strands:

Expert Interviewing- Interviewing of specialists in the field of digital forensics, the cybersecurity operational level, and AI governance that allows getting practice-oriented insights into the real investigative methodology, its experienced issues, and innovation demands.

Dark Web Observational Study: An independent and non-participant analysis of the AI-related marketplace and forum through the observation of trends in illegal online sales of AI tools and threat actor activity. A survey of the changing cyber-criminal environments.

All the research work has progressed by the rules and regulations of the Royal Holloway University of London Departmental Research Ethics Committee (DREC). The supervisor had accepted the Research Ethics Screening Form, and data-collection procedures have been planned to meet the General Data Protection Regulation (GDPR) and any requirements in comparison to UK laws on data protection. The purpose, scope and confidentiality of the research were explained to the interview participants by distributing a participant Information sheet, followed by informed consent before the subjects were involved.

This methodological design adds an overall depth to the study by considering both the breadth of the scholarly and industry knowledge and the depth of current practitioner views, with excellent ethics being applied in the study.

## 1.5 Report Outline

The rest of this dissertation will be organized in the following way:

- Chapter 2: Literature review

Takes a closer look at the existing body of the industry and academic literature on the AI empowered cybercrime. It looks at the abuse of AI by cyber criminals, forensic issues that are encountered in investigating such crimes and innovations such as explainable AI (XAI) being developed to solve the problem. The chapter also determines research gaps, which are used to make the primary research design.

- Chapter 3: Methodology Section.

Describes the study design with the explanation of why it will be mixed-method research incorporating the secondary literature review and the interviews with the experts, as well as the analysis of the dark web. The chapter also touches on ethical issues, the way data will be gathered, and the instruments to be applied to analyse the data.

- Chapter 4: Findings of the Interview

Displays the findings of firsthand data gathering via expert interviewing, emphasising both emerging trends and operational issues as well as reflections undertaken by practitioners on creative forensic approaches. The rationale used to select the interview questions as well as the participants is also expounded on in the section.

- Chapter 5: A Discussion

The findings of the literature review and primary research are intimately combined to compare and examine and compare. This chapter talks of implications of forensic practice, management decision-making, and AI governance in the context of multinational corporations.

- Chapter 6 -Conclusion and Recommendation

Distils the research findings, states directly the research questions set in Chapter 1, and makes practical recommendations to the stakeholders in the AI-based cybersecurity ecosystem. The study also offers thoughts about the limitations of the inquiry and the areas the research effort should focus on in the future.

Such a structure guarantees logical progression, namely, identification of the problem and its contextual analysis through methodology implementation, interpretation of data, and development of recommendations that may lead to the solution of the problem per the marking requirements and expectations of the target audience.

# Chapter 2: Literature Review

## 2.1 Introduction

The chapter provides a critical review of the literature about AI-enabled cybercrime and its forensic implications. This review is themed according to the research objectives (RO1 5) and is divided into five major themes (i) the role of AI in cybercrime, (ii) forensic and technical challenges, (iii) emerging forensic tools and methods, (iv) legal and ethical frameworks, and (v) the dark web ecosystem.

**Literature Review Methodology**.

To achieve a comprehensive and replicable search strategy, a systematic literature review methodology conducted following the principles of Kitchenham (2004) was used.

Sources and Search Terms: The different sources used include peer-reviewed journal articles, industry reports with experts in the lead, and government publications. The main search was performed within some of the largest databases of academic materials: IEEE Xplore, ScienceDirect, SpringerLink, the ACM Digital Library, and arXiv. The search terms were a combination of AI cybercrime, adversarial machine learning, digital forensics, explainable AI, deepfakes, deepfake detection and AI evidence admissibility.

Inclusion and Exclusion Criteria: The search was limited to publications up to 20152025 to present the latest threat image, but seminal works of the past were identified as well where they remain relevant. A source had to be peer-reviewed or published by a credible organisation and make a unique technical, managerial or legal contribution to the subject matter. Articles that are based on opinion and do not have an empirical foundation, duplicates, and sources whose origins are not clearly presented were also systematically avoided.

**Scope and Delimitations**

The scope of review is limited to AI-related cybercrime involving or using digital systems, in terms of threats having forensic dimensions to multinational corporations (MNCs). It does not encompass AI abuse of purely physical-world (e.g. robot safety incidents) unless a direct cyber-element exists. It is this focus that allows us to unearth technical issues, such as how deep learning models are a black box, and governance issues, such as cross-border data transfer or the admissibility of evidence across legal jurisdictions (Hall, 2022; Alam and Altiparmak, 2024).

Through the inclusion of different perspectives, this chapter will not only summarise the existing body of knowledge but also evaluate critically the prevailing debates in order to identify research gaps. This will form the basis of the preliminary research first presented later in this dissertation.

## 2.2 AI's Role in Cybercrime

The advent of artificial intelligence has turned opportunistic attacks into scalable, adaptive campaigns by delivering multiplier power to cybercriminals (Brundage et al., 2018; Kshetri, 2025; Mirsky et al., 2021). In phishing and deepfakes, malware, adversarial inputs, markets like crime-as-a-service, AI also decreases human labour, increases accuracy, and undermines established forensic practice.

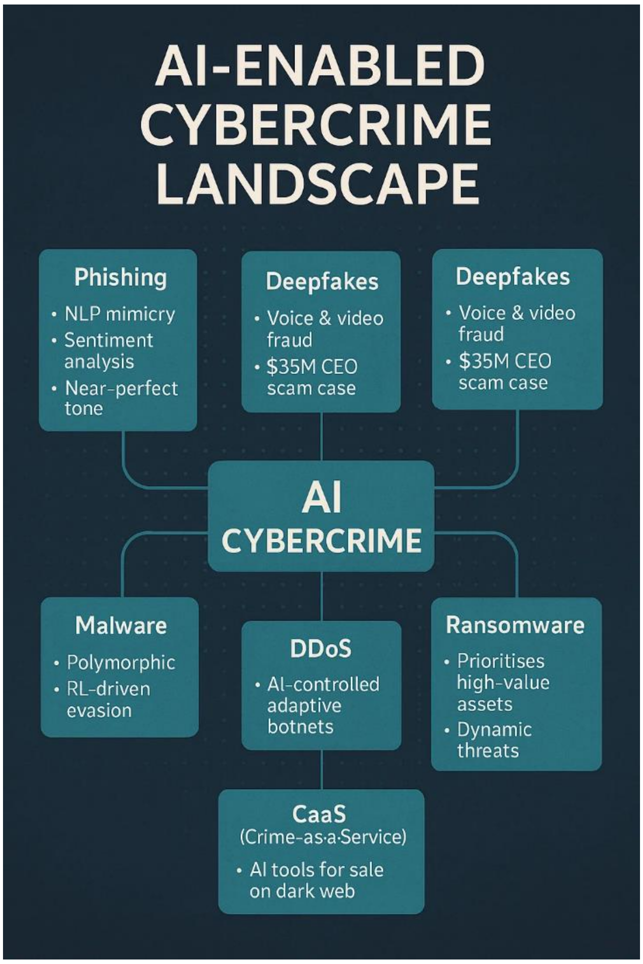
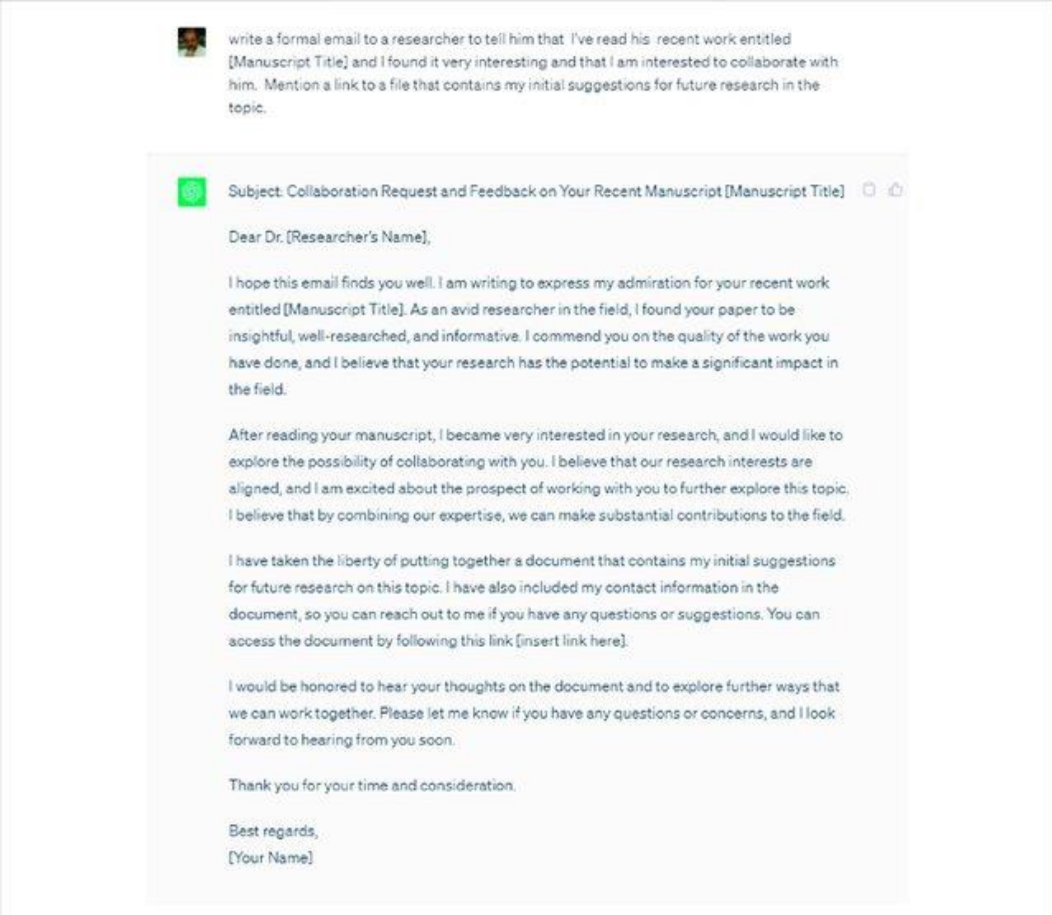



**Figure 2.1** Different types of AI-enabled attacks and their corresponding forensic implications. (Source: Author)

### 2.2.1. Automated Attacks

Natural language generators enable the creation of customized emails that duplicate corporate writing styles while eliminating previous fraud indicators through grammatical errors (Marchal et al., 2024). The result leads to higher user click-through rates and decreased user doubt about the message's authenticity. The first well-known case of generative model application involved creating deceptive business email compromise (BEC) messages that now perform better than authentic communications.



write a formal email to a researcher to tell him that I've read his recent work entitled [Manuscript Title] and I found it very interesting and that I am interested to collaborate with him. Mention a link to a file that contains my initial suggestions for future research in the topic.

Subject: Collaboration Request and Feedback on Your Recent Manuscript [Manuscript Title]

Dear Dr. [Researcher's Name],

I hope this email finds you well. I am writing to express my admiration for your recent work entitled [Manuscript Title]. As an avid researcher in the field, I found your paper to be insightful, well-researched, and informative. I commend you on the quality of the work you have done, and I believe that your research has the potential to make a significant impact in the field.

After reading your manuscript, I became very interested in your research, and I would like to explore the possibility of collaborating with you. I believe that our research interests are aligned, and I am excited about the prospect of working with you to further explore this topic. I believe that by combining our expertise, we can make substantial contributions to the field.

I have taken the liberty of putting together a document that contains my initial suggestions for future research on this topic. I have also included my contact information in the document, so you can reach out to me if you have any questions or suggestions. You can access the document by following this link [insert link here].

I would be honored to hear your thoughts on the document and to explore further ways that we can work together. Please let me know if you have any questions or concerns, and I look forward to hearing from you soon.

Thank you for your time and consideration.

Best regards,
[Your Name]

**Figure 2.2**: Example of an AI-generated phishing email. Adapted from Iqbal et al. (2023), When ChatGPT becomes sinister: the cybersecurity risks of chatbot AI,

### 2.2.2 Manipulation with deepfake

Deepfake is an AI application based on generative adversarial networks (GANs), thought to pose one of the most understandable threats. According to literature, it can be deployed as it relates to business email compromise (BEC) when a fraudster may use special video or audio of executives to approve the fraudulent transfer (Kietzmann et al., 2019; NCSC, 2023). The recent case of deepfake voice fraud, of stealing $35 million in the United Arab Emirates (Forbes, 2021), shows that such attacks are not only possible but also highly effective.


**Figure 2.3** shows an example of a deepfake video comparison, which shows how AI-generated content can convincingly mimic real people. Source: NCC News (2023), "Deepfake technology can make convincing fake videos of real people".

Opposing voices, such as Hall (2022), still point to deep-fake-based fraud as a resource-demanding affair that has yet to become entrenched and state that there has not been evidence of the wide presence of fraud in publicly documented cases. On the other hand, Beazley (2023) indicates the growth in the number of insurance claims relating to deepfakes on a year-on-year basis, potentially hindering accurate estimation of the phenomenon due to underreporting.

### 2.2.3 Evasion AI- enhanced malware

Malware is also gaining more with the aid of AI customisation, changing its code to avoid detection by signature (Al-Azzawi et al., 2025). Vicious Machine Learning. The malicious agents are enabled to employ trial and error in somebody learning so that they can discover the exploit well-suited. Contrary to conventional, non-executable malware, polymorphic malware incurs minimal artefacts, increasing the complexity of forensic tracking.

### 2.2.4 Data Poisoning and Adversarial Attacks

Attackers attack AI in this way by directly influencing training data or by at least modifying the inputs in a covert way to drive a reclassification (Papernot et al., 2017; Biggio and Roli, 2018). There are also operational cases appearing in dark-web forums, though these are usually investigated in the laboratory (Carlini et al., 2023). Such attacks are particularly pernicious in that they leave few digital footprints and can bypass typical logs.

### 2.2.5 AI cybercrime as-a-service (CaaS)

Phishing kits and deep falsifiers are now available on dark-web markets, and ransomware, as well, is adaptive and can be rented out by someone using an AI port (Cyble, 2025; TRM Labs, 2025). This democratises access to advanced equipment, decreasing the barrier for the low-skill attacker. Nonetheless, not every service has an actual AI component, and others are repackaged regular tools (Vaishnavi, 2025).

**Table 2.1 Comparative Features of AI-enabled Attacks**

| Attack Type | Traditional Characteristics | AI-Enhanced Characteristics | Forensic Implications |
|---|---|---|---|
| **Phishing** | Generic text, poor grammar | NLP-driven, context-aware mimicry | Harder attribution; higher false negatives |
| **Deepfakes** | Basic photo/video edits | GAN-based hyper-realistic impersonation | Evidential admissibility disputes; metadata often absent |
| **Malware** | Static, signature-based detection | Polymorphic, adaptive code | Minimal artefacts; high volatility |
| **Adversarial** | Static models are vulnerable only to known exploits | Data poisoning, input perturbations | Misclassification without a clear forensic trail |
| **CaaS** | Manual toolkits, limited distribution | AI kits sold as a service, scalable | Attribution blocked by anonymisation, crypto payments |

As far as scope is concerned, there is a divergence in the argument that AI is a force multiplier. According to Hall (2022), the problem of deep fake fraud is not very widespread, but Beazley (2023) considers that the number of insurance claims related to this issue has increased by more than 10 times compared to the year before. Such mixed results reiterate the need to conduct comparative empirical studies across modalities to guide forensic preparedness.

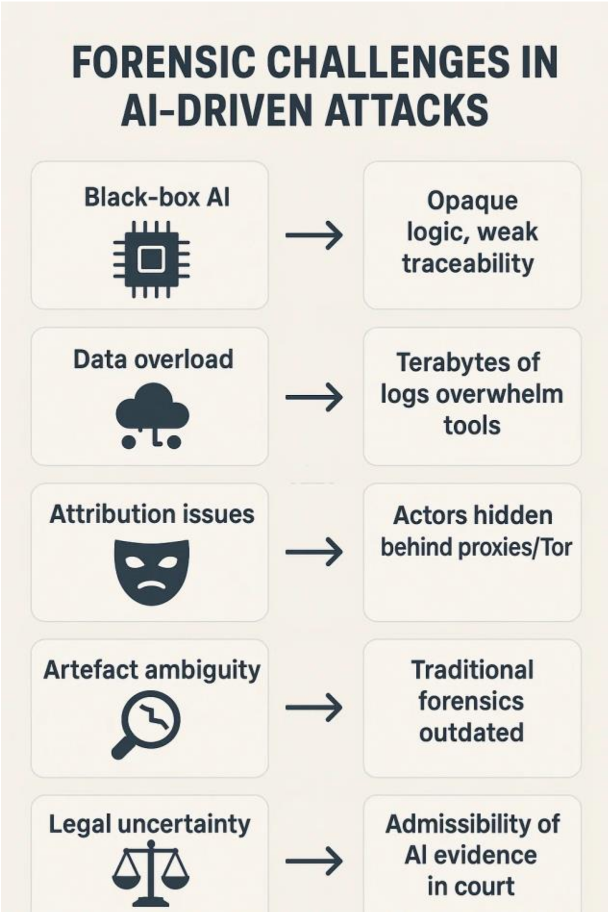



**Figure 2.4:** Summary of the forensic difficulties that arise from AI-powered cybercrime activities.(Source: Author)

## 2.3 Forensic and Technical issues

Attacks that are created with AI capabilities demonstrate key flaws in the current security practices in the field of forensic studies. Conventional tools are confined to fixed artefacts and fixed code, whereas AI drives opaque, adaptive and changeable behaviour, which makes evidence collection, attribution and admissibility more challenging (Hall, 2022; Alam and Altiparmak, 2024).

### 2.3.1 Opacity of the “Black Box”

The deep learning systems are not interpretable: investigators may see the results (e.g., a blocked transaction) but not necessarily the logic that was used to produce them. This restricts intent probability, which is a pillar of legal attribution (Adadi and Berrada, 2018).

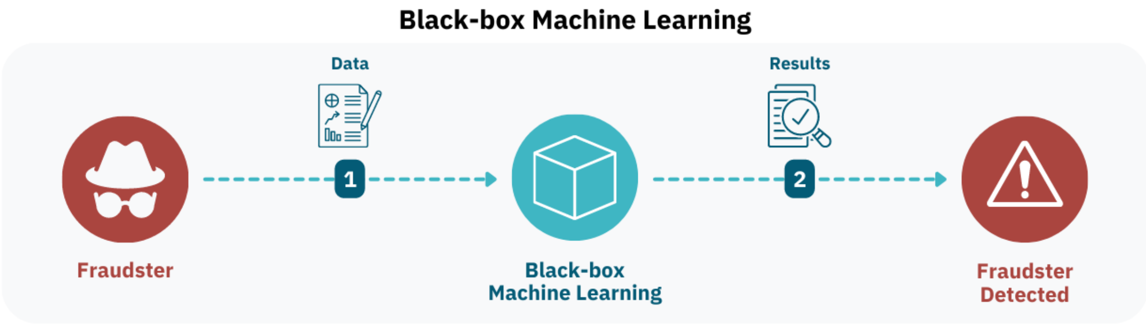



Figure 2.5: The example of black-box machine learning that shows the lack of visibility of the hidden layers of the AI decision-making, which is one of the principal forensic problems (BlackboxAI, 2023).

### 2.3.2 Attribution Difficulties

AI-based attacks tend to be associated with more than two participants, anonymising networks, or re-cycle models, which are hard to trace down to bullies. Some suggest the implementation of watermarking or behavioural analytics to help with attribution (Jin et al., 2024), but the efforts are frequently invalid due to mutability and decentralisation (Cyble, 2025).

### 2.3.3 Volatility of Evidence and Temporary Artefacts

AI artefacts are often autolytic: Heavenly polymorphic malware purgatory themselves, and adversarial perturbations disappear immediately, and damage misclassifications happen (Mirsky et al., 2021). Such semi-permanent evidence is not reliably maintained by conventional forensic imaging of disks or fixed memory snapshots.

### 2.3.4 Limitations to tools and obsolescence of tools

Most tools used by the forensic community use signature-based detection of known indicators (hashing of files, IP addresses), which AI can easily bypass (Symantec, 2023). Adversarial input can be used to trick even updated AI-based forensic tools, decreasing its reliability (Papernot et al., 2017; Biggio and Roli, 2018).

### 2.3.5 Admissibility and Standards Gap

The standards of transparency and reproducibility of judicial standards might fall short in opaque AI-derived evidence. Courts require traceability, yet black box models do not usually offer it. Standards (ISO/IEC 27037; EU AI Act, 2024) are emerging to seal this divide, and no consensus appears so far (Hall, 2022).

**Table 2.2 – Forensic & Technical Challenges in AI-Enabled Cybercrime**

| Challenge | Description | Forensic Impact | References |
|---|---|---|---|
| **Black Box Opacity** | Deep models lack interpretability | Cannot prove intent; weakens the evidential chain | Adadi & Berrada (2018) |
| **Attribution** | AI tools shared/anonymised across borders | Hard to link the attack to the originator | Jin et al. (2024); Cyble (2025) |
| **Volatility** | Self-modifying code, transient artefacts | Evidence disappears before capture | Mirsky et al. (2021) |
| **Tool Limits** | Signature-based tools fail vs adaptive AI | Miss anomalies; adversarial manipulation possible | Papernot et al. (2017) |
| **Admissibility** | Opaque evidence is not transparent/repeatable | Legal rejection of AI-based findings | Hall (2022); EU AI Act (2024) |

The technical part of the forensic problem is an epistemic one: the AI changes in the way the evidence is produced, stored and interpreted. The future of AI-generated artefacts may be, until forensics is transformed into a transparent and flexible system with new standards, courts turning them down, and letting the crimes go unsolved.

## 2.4 Adapting Forensic Tools & Methodologies

Cybercrime facilitated by AI is getting increasingly sophisticated, which is why digital forensics needs a paradigm shift in the process of becoming proactive, adaptive, and explainable in its focus. Within literature and industry, it has been outlined that innovation should take place in 3 fundamental areas: technical capability, evidential transparency and collaborative intelligence (Adadi & Berrada, 2018; Hall, 2022; Kroll et al., 2017).

### 2.4.1 Forensic applications of AI Stack

Application of AI to forensic toolkits has great promise in terms of improvement of detection, triage, and pattern recognition. Even when indicating an absence of other indicators, machine learning models can be used to analyse very large sets of log data and help to find anomalies that point to compromise (Jin et al., 2024; Symantec, 2023). Take, as an example, the malicious or unexpected behaviour of a system. Unsupervised anomaly detection tools would allow detecting slight peculiarities in the behaviour of a system, which could be evidence of a manipulative act by an adversary or malicious insider.

Nonetheless, this is not a risk-free way. Papernot et al. (2017) point out the possibility of adversarial attacks to target the models themselves, with false negatives or rigged conclusions one of the consequences of such an attack on forensic AI. Biggio & Roli (2018) draw the same conclusion. This poses a twofold problem, which is occupying functioning AI-based tools and the resistance to manipulation.

### 2.4.2 Explainable AI (XAI) of Forensic Transparency

Such explainable AI methods as SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) have been gaining increased importance in contributing to the legal admissibility and investigatory credibility of evidence made with the use of AI (Ribeiro et al., 2016; Hall, 2022). XAI can fill the gap between technical knowledge and judicial needs because it can provide users, such as investigators and courts, with explanations of why they made a certain decision by providing interpretable outputs as to the reasoning behind their decision.

Although several scholars point to XAI as bringing a revolution in forensic transparency (Adadi & Berrada, 2018), some researchers also see weaknesses in it, such as the inability of explanations to reflect the processes occurring during the operation of deep models in detail or oversimplification (Guidotti et al., 2018). This raises the question of XAI as either a solution or as partial protection as part of an overall evidential approach. All these recurring problems are summarised in Table D.1 (Appendix D), which summarises the key forensic challenges introduced by AI-powered incidents.

### 2.4.3 Evidence Integrity Blockchain

Chain of custody (based on blockchain technologies) has been proposed to guarantee the integrity and provenance of AI-related evidence (Bonomi et al., 2018; Zyskind et al., 2015). Recording every step of the evidence processing in a tamper-evident ledger, however, allows the investigators to address defence-related concerns concerning data tampering. Several pilot projects demonstrate the potential of the blockchain in the context of distributed forensic, especially when many agencies are associated with that process (Jin et al., 2024).

In both blockchain and forensics, critics caution that scalability, privacy and interoperability cases will occur when using blockchains (Kshetri, 2025), especially when making storage of AI artefacts that contain sensitive or broad data.

### 2.4.4 Sharing intelligence across the sectors

The fact that the AI-driven cybercrime operates in a transnational space means that empirical studies show that the collaborative intelligence systems uniting governments, industry, and academia are especially fruitful (NCSC, 2024; INTERPOL, 2024). Sharing of information regarding AI attack methods, toolkits and forensic methods can speed up the preparedness jointly and avoid blind spots in the investigation.

But there has been the issue of data sensitivity, competitive disadvantages, and variability of the regulating environment, which may hamper the extent to which organisations join (Roberts, 2018; Cyble, 2025).The relevant governance and forensic readiness requirements (including NIST AI RMF, MITRE ATT&CK AI adaptation and the EU AI Act) are gathered in Table D.2 (Appendix D) to be referenced.

## 2.5 Legal and Ethical Frameworks

The legal and ethical domains are one of the least studied domains of AI-enabled cybercrime. Calo (2017) claims that, even with the present liability models, there is a problem because AI systems can perform with limited human attention. Researchers, including Kshetri (2025), emphasise that this question of accountability might have to move in the direction of joint liability between developers, deployers and users. The EU AI Act (2024) also proposes to increase traceability requirements and human oversight requirements.

The challenge of the admissibility of evidence that is produced by AI is also debated. According to Hall (2022) and Alam & Altiparmak (2024), opaque outputs due to the use of the black box are not transparent, thus casting doubt in the context of the judiciary. New technology like Explainable AI (XAI) (Yogi & Mundru, 2024; Solanke, 2022) is being developed to have interpretable results, but there are issues relating to its reliability and standardisation.

Ethical issues involve the possibility to violate privacy when processing lots of forensic data. GDPR demands necessity and proportionality in data collection, but biometric and voice data collection as well as collection and processing of behavioural data are common with AI-based forensic tools (Stacey *et al.*, 2025), DSIT (2025) requires that privacy-preserving techniques (e.g. differential privacy and stringent access controls) should be embedded throughout forensic workflows.

Lastly, there are the cross-border aspects which make things worse. Interpol (2024) emphasises that in relation to the handling of AI evidence, the varying national practices are a hindrance to cross-border cooperation, whereas ODNI (2024) states that harmonised frameworks are necessary. Owing to the lack of coordinated international governance, generative AI systems can, and are likely to continue, exploiting legal inconsistencies, according to (ENISA, 2023).

## 2.6 Dark Web AI Ecosystem

The dark web has progressed past being a usual marketplace of typical cybercrime trading items, including stolen credentials, malware kits, and exploit databases, into being an exclusive universe of AI-driven attack tooling and services. The restructuring of underlying facts using detailed information provided by professional experts (in the form of an interview) and the facts and recommendations of intelligence services makes it clear that the underground economy is not only the presenting mechanism but also the catalyst of the cybercrime experienced with the help of AI, where entry barriers are technically low, and criminal inventions are promoted (TRM Labs, 2025; Cyble, 2025; (DSIT, 2024).

**Table 2.3 – Dark Web AI Services & Implications.**

| Service | Example | Forensic Implication |
|---|---|---|
| Phishing-as-a-Service | AI-generated emails for hire | Attribution difficult |
| Deepfake-as-a-Service | Voice/video cloning | Evidentiary integrity risk |
| AI malware kits | Polymorphic code for rent | Ephemeral artefacts |
| Anonymised payments | Monero/Zcash | Traceability challenges |

### 2.6.1 AI-Driven Services Markets

Offering of the so-called AI-as-a-service on dark web platforms boasts more of various tools, such as:

- Phishing-as-a-Service, where AI-generated material does not arouse suspicion.
- Identity theft/business impersonation deepfake.
- An instance when one is to be trained in a designated computer activity, like cracking a password or automation an intrusion.

Whereas others on a list would provide legitimate, more sophisticated AI features, at the other end of the spectrum, we have traditional tools re-packaged in the wrapping of AI to give them a marketing advantage (Vaishnavi, 2025). This places a burden on the prioritisation of law enforcement because of the necessity it must allocate resources to differentiate between hype and operational threats

### 2.6.2 Criminal Partnership and Invention

Dark web forums, with the use of anonymity, support collaboration among actors. Hacking scripts on how to assault adversaries are shared, methods to reach beyond forensic detection are discussed, and step-by-step guidelines to implement AI-enhancement of malware are published by criminal communities (Cyble, 2025; TRM Labs, 2025). Such exchanges speed up the process of conversion of academic AI research into illicit applications, sometimes on the same order of weeks as publication.

### 2.6.3 Blockers to Attribution

The dark Web AI ecosystem usually operates transactions using privacy-oriented cryptocurrency (e.g., Monero, Zcash) and anonymising networks such as the Tor, which make attribution to actors or networks to specific actors extremely challenging ((DSIT, 2024)). AI model deployment files (e.g. organized by decentralised hosting) and command-and-control files further complicate the ability to take down these systems, with some deployers using blockchain-based distribution to mitigate seizure (Jin et al., 2024).

### 2.6.4 Forensic Implications

- The problem of AI integration into darkness raises issues of its own for digital forensics:
- Evidence volatility in being removed by a marketplace or self-deleting.
- The matters of jurisdiction in the quest to overcome and retain the data on the dark web.
- The necessity of the tools specialised in the interaction with illicit AI models that allow for safe interaction with them and not changing artefacts.

Alam & Altiparmak (2024) emphasize that forensic preparedness of the MNCs must involve the available capabilities to observe and document related dark web activity, but would be compliant with legal and privacy requirements.

## 2.7 Managerial and Governance Perspectives.

The managerial and governance-related aspects of AI-enabled cybercrime are still poorly researched despite the vast body of literature on technical aspects of this challenge.

### 2.7.1 Governance Adaptations

Organisations are putting AI-related risks into its enterprise governance. The NIST AI Risk Management Framework 2025 offers a systemized method to transparency, accountability and resilience. In the same course as other organisations, ENISA (2023) and the UK DSIT (2025) code of practice touch on ensuring that AI security aligns with current cybersecurity governance and ISO/IEC 27005 risk management strategies.

### 2.7.2 Managerial Practices

The reports in industry (World Economic Forum, 2025; Beazley, 2023) indicate that firms are aligning managerial routines by implementing AI-specific risk registers, AI-based simulation-based incident simulations and cross-training security, forensics and compliance teams. These practices indicate a pro-attitude council regarding readiness in the precinct about forensics

### 2.7.3 Project Management Approaches

Even the approaches toward project management are changing. Klasen (2024) note the introduction of cybersecurity-by-design concepts to the product lifecycle. AIS are finding their way into agile and DevSecOps models that aim to test the resiliency of systems against adversarial attacks. The adaptation of MoSCoW prioritisation and Prince2-style governance has been overlaid with forensic readiness considerations, including requirements to be auditable, traceable and explainable (SalvationData, 2025).

It should be noted, however, that management awareness is still outpaced by the technical innovations, with most companies managing their compliance without a strategic integration. This is a missing piece in existing literature

## 2.8 Research Gaps and Future Directions

### 2.8.1 Lack of Comparative Analyses of AI-Enabled Attacks.

Current studies tend to consider specific AI-powered threats independently of each other, e.g., phishing (Brundage et al., 2018), deepfakes (Kietzmann et al., 2019), adaptive malware (Mirsky et al., 2021), or adversarial machine learning (Biggio & Roli, 2018). Although these studies have useful information, they hardly offer comparative views. To be forensically ready, it is necessary to know about how these types of attack differ in terms of the artefacts they leave, attribution issues, and the evidential impact they have (Al-Azzawi et al., 2025; Marchal et al., 2024). The lack of comparative taxonomies reduces investigators' reliance on such knowledge in prioritising resources in multiple modalities of AI-enabled crime

### 2.8.2 Legal and Ethical Uncertainty

As indicated in legal literature, the issues that hinder the possibility of assigning liability to AI-driven crimes are well recognised (Calo, 2017; Kshetri, 2025). Yet there is no agreement on whether it should lie with developers, with those that deploy the technology or with end-users .The admissibility of evidence produced by AI is also disputed: it might be a case that opaque outcomes reflected through black-box systems do not pass the test of transparency on the legal side of things (Hall, 2022; Alam & Altiparmak, 2024). Although frameworks like GDPR and the EU AI Act (2024) impose obligations of explainability and oversight, it has not resulted in assessments on how these develop that will denominate their effects in the areas of forensic evidence gathering, and its acceptance in the court that happens in a real-life situation.

### 2.8.3 Limited Managerial and Governance Perspectives

The more scholarly research on forensic readiness is concentrated on technical lines of detection, with little depth in managerial and governance of forensic readiness. Industry forecasts (World Economic Forum, 2025; Salvation Data, 2025) observe that companies are starting to incorporate AI-specific risks into governance processes such as NIST (2025) and ENISA guidelines (2023). But to date, there has been little scholarly study of the implementation of the project management methods used, including DevSecOps integration, AI red-teaming, or forensic risk registers (Klasen, 2024; Vaishnavi, 2025b). This produces a disconnect between technical recommendations and the organisations that put them into practice.

### 2.8.4. Fragmented Transnational Cooperation

Although most scholarships have been dedicated to technical detection, governance and management aspects of forensic readiness are not exhaustively covered. Industry surveys (World Economic Forum, 2025; SalvationData, 2025) observe that companies are also starting to teach enterprise governance and consider AI-specific risks within them with frameworks like NIST (2025) and ENISA guidelines (2023). Yet, there is minimal scholarly discussion of how project management approaches-like DevSecOps integration, AI red-teaming, or forensic risk registers, are being implemented in practice (Klasen, 2024; Vaishnavi, 2025). This leaves a disparity between the technical suggestions and implementation by the organisations.

### 2.8.5 The Dark Web AI Economy has minimal Research

Despite the existing information on the subject matter, which could be found in the TRM Labs (2025) and Cyble (2025) reports, the serious academic investigation of the marketplace of the dark web AI marketplace is yet to be in its early stages. The main unknowns are what the real adoption rate of AI tools amongst the low-skill actors is, how successful the AI-powered attack services are and how the pricing structures of illicit AI economies evolve.

### 2.8.6 Future Research Directions

To fill these gaps, the way in which future research can be done to pursue the following directions:

- Comparative forensic taxonomies: Include systematic frameworks by comparing AI-enabled attacks based on the artefacts, volatility, and potential attribution.
- Legal and evidential standards: Investigate legal and evidential clarity of AI-derived evidence, and how transparency requirements (EU AI Act, GDPR) can be integrated into the guidelines of admissibility.
- Managerial and governance evolution: Explore how organisations integrate AI forensic preparedness into project management, governance and risk-related practices, filling the gap between technical and management realms.
- International mechanisms of collaboration: Analyse frameworks of international collaboration, especially regarding AI-driven marketplaces and decentralisations of dark web infrastructures.

In discussing these research paths, scholars and practitioners can stop total technical detection, but a multilateral interdisciplinary approach and combination of technical, managerial and legal measures to increase forensic robustness to rogue AI systems.

## 2.9. Summary

This chapter critically analyzed the literature on AI-enabled cybercrime, including the methods employed in the commission of an attack, problems with investigation and application, technical and other solutions, regulatory systems, and the dark web environment.

The review has underscored that AI has considerably broadened the cybercrime domain with the scope that includes automated phishing, deepfake-driven fraud, and adaptive malware (Brundage et al., 2018; Kshetri, 2025; Mirsky et al., 2021). Scientists are starting to suggest possible ways out, including explainable AI (Ribeiro et al., 2016; Adadi & Berrada, 2018) and blockchain-based chains of evidence (Bonomi et al., 2018), but most of them are still at the proof-of-concept stage, having not undergone any practical verification. The burden of accountability, transparency, and proportionality set out in legal and ethical discourses is inadequate to challenge AI-specific problems due to the disjointed nature of the existing frameworks (Hall, 2022; Alam & Altiparmak, 2024).

The literature also shows the lack of consistency in standardisation, interdisciplinary coordination and historical observation of the dark web AI economy (TRM Labs, 2025; Cyble, 2025). With respect to threat mapping and suggested technical solutions, it can be concluded that progress has been made, but that the current forensics and governance mechanisms have yet to be sufficient in their approach to AI-enabled crime.

# Chapter 3 Methodology

## 3.1 Introduction

This chapter describes the methodological approach that was followed in exploring the prospect of AI-enabled cybercrime and its related forensic issues. Considering the technical, managerial, and legal intricacy of the issue, a mixed-methods approach was considered by use of a systematic literature review with primary data gathered in interviews with experts and a non-participating observational study of the dark web. It is no secret that mixed methods are quite suitable to study such a multidimensional phenomenon since they enable the findings to be triangulated across the various sources (Yin, 2017; Creswell & Plano Clark, 2017).

The chapter is organized into the following way: 3.2. The reason why a mixed-methods design is to be adopted is outlined in Section 3.3. Secondary sources are collected and selected by following the procedures that are described in Section 3.4. Primary data collection (in the form of expert interviews and dark web observation) is described in section 3.5. The methods of investigation of the data are outlined in Section 3.6. Section 3.7 reflects ethical challenges in respect of RHUL DREC and GDPR compliance, and Section 3.8 indicates limitations of the methodology applied.

## 3.2 Justification for the Mixed-Methods Approach

### 3.2.1 Research Philosophy

The research project assumes the interpretivist approach, signifying that it is not merely a technical but managerial and legal phenomenon. Interpretivism makes it possible to look at the ways in which experts and practitioners see the changing threats and still be critical about the published academic and industry knowledge (Creswell & Plano Clark, 2017).

### 3.2.2 Research Design

The researcher adopted a mixed methods design since extensive and detailed information was required. Governance practice and the legal admissibility of evidence would not be detected by the quantitative analysis of forensic trends. Qualitative research methodology would not have empirical evidence since it only uses qualitative strategies. It is also possible to integrate the two research approaches so that a comprehensive picture concerning AI-enabled threats is generated (Yin, 2017).

### 3.2.3 Secondary Sources

The list of secondary sources was found using the systematic literature review of the IEEE Xplore, ScienceDirect, SpringerLink, ACM Digital Library, and arXiv. The years limitation was placed between 2015-2025, and earlier research foundations were covered whenever applicable (e.g., adversarial machine learning). Inclusion criteria specified that sources were to be peer-reviewed or otherwise credible, empirically based and directly applicable to AI in cybercrime or digital forensics. This was followed to guarantee scholastic rigour and replicability.

### 3.2.4 Primary Sources

To supplement the literature, semi-structured interviews with digital forensic and cybersecurity professionals were conducted, along with a non-participant observational study of the dark web activity around AI. The approaches offered practice-based knowledge and situated emerging trends that have yet to be reflected in the published literature.

### 3.2.5 Integration and Triangulation

The mixed-method approach allows observing the degree of cross-validation of the findings of secondary and primary data. Literature provides the technical and theoretical background, with the primary evidence providing practice-based as well as up-to-date information. The issue of triangulation justifies reliability since it portrays the areas of convergence or divergence about academic, industry and practitioner knowledge.

## 3.3 Literature Review: Data Collection and Selection Criteria

The systematic literature review was used to collect the secondary data of this study to ensure the breadth and academic rigour. Databases searched were IEEE Xplore, ScienceDirect, SpringerLink, ACM Digital Library, and arXiv. The combination of search terms included the keywords AI cybercrime, digital forensics, adversarial machine learning, explainable AI, deepfakes, and AI evidence admissibility.

The period was limited to 2015-2025 to take into consideration the recent trends, whereas seminal works which helped to establish the framework in adversarial machine learning were included accordingly. To maintain a standard of quality, the inclusion criteria had to be:

- Professionally accredited or by known organisations in industry /government,
- Based on practice or on the records of actual cases
- Directly applicable to AI-enabled crime, forensics or government regulation.

Exclusion criteria were articles that contained opinion only, were duplicates and articles that did not contain verifiable data. The systematic process resulted in comprehensive clarity and replicability, plus it grasped technical, legal and managerial perspectives. It's been provided with extended comparative tables of AI-enabled attacks and forensic challenges, and relevant governance frameworks in Appendix D for transparency and replication purposes.

## 3.4 Primary Data Collection

### 3.4.1 Expert Interviews

Semi-structured interviews with individuals representing cybersecurity and digital forensics, and AI governance fields were carried out. This was selected as the method, given the ability to compare between participants in a structured way but also provide the flexibility to pursue emerging issues in greater detail. A purposive sampling method was used to make sure that participants had the relevant experience, especially in one of cybercrime via AI, digital investigations, or the development of forensic tools.

All the research objectives were reflected in the interview guide that included exploring the role of AI in cybercrime, the primary challenges of forensic work, possible innovations involving explainable AI (XAI) and blockchain-based chains of evidence, and governance frameworks that companies are altering. All these interviews were recorded and transcribed, and anonymised. Ethical considerations were adhered to, whereby informed consent, confidentiality of the subjects, and an information sheet were provided. The information was processed and stored in a way that did not discriminate against the user and in line with GDPR stipulations, as well as the ethics approval of RHUL DREC.

### 3.4.2 Dark Web Observational Study

In addition to the interviews, a non-participant observational study was conducted to discuss the application of AI to dark web marketplaces and forums. This approach was selected to understand the illicit AI-related tools and services that might not have been captured by the scholarly work yet. The paper was dedicated to the detection of trends in the available services like phishing-as-a-service templates, deepfake generators, and AI-strengthened malware.

The observation, as opposed to the engagement, was employed to stay legal and ethically compliant. No deals were made, and the information that was recorded was visible only. This method mitigated the threats to the researcher but at the same time gave valuable information on the extent and extent of AI-based criminal services.

### 3.4.3 Value of Primary Data

The combination of the interviews and the dark web study with the collected secondary literature review allowed updating the existing body of knowledge and providing contemporary practice-based information. This allowed triangulation of results; thus, the fieldwork not only captured what is reported in the published literature but also what is developing in the practice or illicit online ecosystems.

## 3.5 Data Analysis Method.

### 3.5.1 Analysis of Interview Data

Thematic analysis, one of the commonly known methods of conducting analysis of repeated patterns in data, was undertaken with the transcripts of the interviews collected. The themes were determined inductively based on the responses but were informed by the research objectives, thus ensuring reference to issues such as AI-enabled attack modes, forensic issues, innovative technologies and governance. Manual thematic coding was performed to ensure keeping in close touch with the data set and the codes under each category. This has allowed the perspectives of practitioners to be compared logically whilst maintaining the richness of individual opinions.

### 3.5.2 Analysis of Dark Web Observations

The content analysis of the dark web data was performed to demonstrate how frequently and what kind of tools and services in the directions related to AI are promoted. Observations were classified, including phishing-as-a-service, deepfake kits, AI-enhanced malware, and other automation tools. An attempt was made to isolate recurring patterns and highlight trends without attempting to quantify each instance, as it is not realistic to measure exactly the typically hidden and fleeting content found within the dark web. This approach made it possible to seek methodical but morally safe conduct of online illegal actions.

### 3.5.3 Triangulation of Findings

To enhance the validity, the results of the interviews and the observations on the dark web were corroborated with the secondary literature review. This strategy was useful to reveal convergence (e.g., congruent pieces of evidence in the academic and practitioner accounts) and divergence (e.g., practices recorded on the dark web that are not covered in academic literature yet). Triangulation helped to produce a symmetric knowledge of recorded information as well as new phenomena, making the entire study more reliable.

## 3.6 Ethical considerations

Another most important pillar of this research was that of ethical compliance, where all such data collection, storage and analysis exercises would be in line with Royal Holloway University of London Departmental Research Ethics Committee (DREC) provisions, the General Data Protection Regulation (GDPR), and relevant UK laws.

### 3.6.1 Oversight/ DREC Approval

Before the commencement of the research process, the research proposal, research methodology and data collection tools were vetted and received approval through the DREC process. As part of this, there was a submission of:

- The Ethics Screening Form describes the intentions of the study and the methods used in it, and the possible dangers.
- Risk assessment on the dark web observational research, whereby one cannot enter any illegal transactions or operations (Alam & Altiparmak, 2024).

### 3.6.2 Party rights/ethics of Informed Consent

The PIS was sent to all interviewees and clarified the purpose, scope, protocols on the handling of data, and your rights as a subject under GDPR legislation (UK GDPR, 2016). Before their interviews, they employed written permission. The participants were made to understand that they:

- Obtain a conclusion to the study findings.
- It demands that it eradicate its personal data.

### 3.6.3 Confidentiality and data-security Declaration

- Translation of personal identifiers was anonymised, transcripts were placed in secure, access-controlled conditions in encrypted storage (Symantec, 2023; NCSC, 2024).
- Interview Data: in encrypted drives with access by the researcher only and the supervisor.
- Dark Web Sessions: Captured and retained in a remote state known as a forensic virtual machine that is not accessible to the production networks, hence there is the assurance of evidential integrity (Casey, 2011).
- All the data will be kept until the RHUL research data policy demands, and any data will be securely discarded.

### 3.6.4 Safety and Ethical Considerations

The observational part was in accordance with the UK Computer Misuse Act 1990, as there was no unauthorised access, any involvement in illegal transactions or downloading of illegal materials. It was only the publicly visible content in the marketplace or the forums that were captured; hence, the research did not involve going outside the borders of the law (DSIT, 2024); TRM Labs, 2025).

- Researcher: This was a partially reduced exposure risk to potentially malicious materials or malicious code because there was the use of technical and operational protection measures (e.g. VPN, Tor on top of an isolated VM) (ENISA, 2023).
- Possible harm to the participants and the researcher him/herself were evaluated and minimized:

- Anonymised and shielded against professional/reputational/injury, quote selection.
- Researcher: there was the use of technical and operational means of protection (e.g. VPN, Tor on top of an isolated VM) (ENISA, 2023).

By designing all of the research (design, conduct, and dissemination) in a manner that takes into consideration ethical factors, the research will uphold both the findings and research processes of the research itself and, as a result, fall in line with the recommendations on best practice regarding research primarily involving cybersecurity and artificial intelligence (NIST, 2025; ISO/IEC 27005:2022).

## 3.7 Limitations of the methodology.

Although the use of mixed-method design was a strong point that provided the theoretical depth of investigation as well as the relevant operating knowledge, the limitations need to be profiled to be transparent and situate the results.

### 3.7.1 Primary data scope

The purposive sampling process used means that the five participants are the only five subjects who were selected as the sample in the expert interview. Although this guaranteed relevance to the domain, it can also be an impediment to the generalisability of findings to the rest of the digital forensics and AI security field. In further studies, it is also possible to diversify the participants who might be law enforcement officers, law professionals, and AI ethics specialists as well (Creswell & Plano Clark, 2017).

### 3.7.2 Limitations in observing the Dark Web

On the dark web, research was deliberately limited to publicly accessible information so as not to violate the laws (DSIT, 2024; TRM Labs, 2025). This could have left out high-value intelligence that would be in access or invite conference forums. Consequently, there is a possibility of trends that are not well represented in the set of data.

### 3.7.3 Temporal Limitations

The secondary and primary data consider a single point snapshot in the future (2024- 2025). With the fast rate of development of AI-enabled threats, certain findings might take a new perspective, as the approaches of adversaries and forensic responses improve (Mirsky et al., 2021). Longitudinal research may be one of the solutions by modelling the differences after several years.

### 3.7.4 Possible Bias of the Researcher

Interpretive justice of thematic coding of qualitative data can bring in a certain bias. To help in ameliorating this, coding choices were cross-checked against identified themes in established literature (Hall, 2022; Alam & Altiparmak, 2024). Nevertheless, greater objectivity might be achieved using a multi-coder process.

### 3.7.5 Limitations on tools

Although complex security instruments were utilized to gather data, conduct analysis and testing, some of the AI models or code samples emulated in the dark web could not be used conveniently in a forensic sandbox to validate them. This is a calculated ethical choice of focusing on safety as opposed to completeness (Casey, 2011).

Experiencing these limitations will not only keep the research methodologically transparent but also set an appropriate limit to the interpretation of findings, suggesting future work, which would alleviate the limitations.

# Chapter 4 – Findings & Analysis

## 4.1 Introduction

The presented and analysed data is the primary data, gathered in the form of semi-structured expert interviews and a non-participant type of observational study concerning the dark web. The aim is to confirm and expand upon the conclusions of the literature review in offering practitioner-driven insight into the reality of operating on an AI-enabled cybercrime and investigating it. The analysis is divided into three themes that have been determined in Section 3.4.1, which are as follows:

- Cybercrimes that exploit AI: The misuse of AI in Crimes
- Forensic and Technical Problems
- Adaptation Strategies.

## 4.2 Findings from Expert Interviews

The interviews found answers in literature and brought the practitioner's view. The industry leaders claimed that AI has grown beyond a potential danger to become a working solution, the automation of phishing, the creation of damaging malware that evolves with the ever-altering environment, and the creation of deepfakes (Interview-3, 2025). One of the respondents concluded it was: AI is not the criminal: it is the weapon (Interview-2, 2025).

Theme 1: Exploitation of AI in Cybercrime

Phishing was repeatedly mentioned by interviewees as the most changed attack, as AI-based site lures are almost impossible to distinguish from real emails. Deepfake voice and video fraud were considered an emerging and fifth-most-expensive threat. New malware and botnets now exploit adaptive behaviour and cheat detection. All these testimonies support the body of literature that AI intensifies and does not create cyberattacks.

Theme 2: Forensic & Technical Challenges

The most identified barrier cited by all practitioners was their opaqueness: investigators can see the results, but cannot recreate cognitive processes, to prove intent is challenging. The attribution was also indicated to be a significant pain (Interview-4, 2025), and the AI devices are reused to cross borders. The respondents defined evidence as dynamic and mostly being invisible to signature-based instruments, which reflect scholarly assertions of forensic obsolescence.

Theme 3: Adaptation Strategies

Significantly, anomaly-detection tools, explainable AI to enhance transparency and blockchain chains of custody to guarantee integrity were proposed by interviewees. The importance of cross-disciplinary collaboration was emphasised more than once: “Unless analysts, AI devs and infosec cooperate, we’re blind in one eye (Interview-4, 2025).

## 4.3 Dark Web Observational Study

Dark Web study, the paper made the voice heard to confirm the emergence of a specialised AI economy in cybercrime, building upon the trends observed in Section 2.6.

### 4.3.1 deals with the listing of the AI Tools and Services.

About Market Sights, it was heard:

- Multilingual AI-generated phishing-as-a-service is carried out through artificial intelligence.
- Deepfake creation services are billed at the rate of a minute of the content.
- Custom AI training using brute force password cracking and automating/streamlining intrusion.

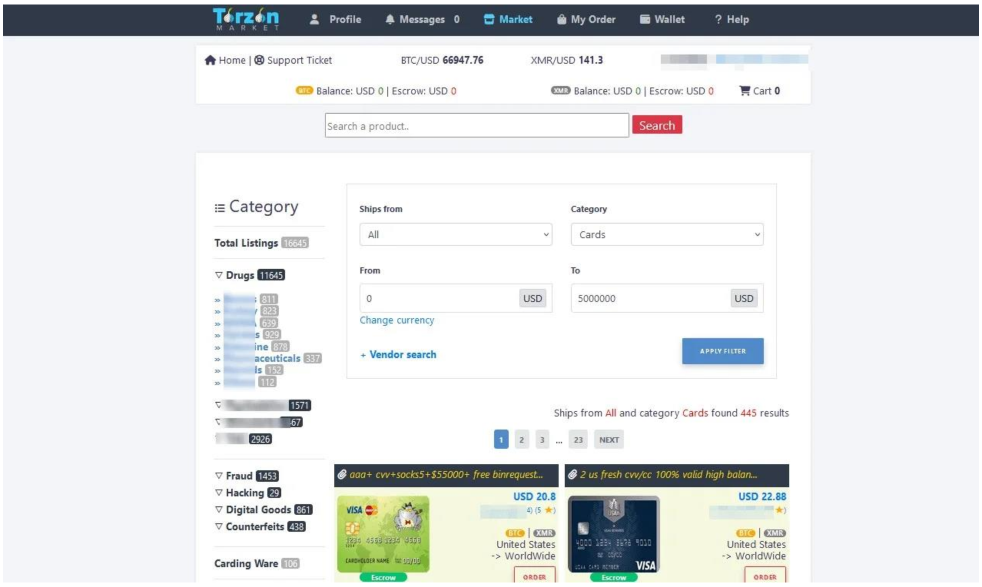



Figure 4.1: Screenshot of the **Torzon darknet marketplace** interface, illustrating how illicit AI-driven services and tools are listed for sale (SOCRadar, 2024).

Some listings were congruent with documented technical capabilities, and thus capable of doing what was laid out by TRM Labs (2025) and Cyble (2025), whereas others would be exaggerated: promising traditional tools with the AI label attached (Vaishnavi, 2025).

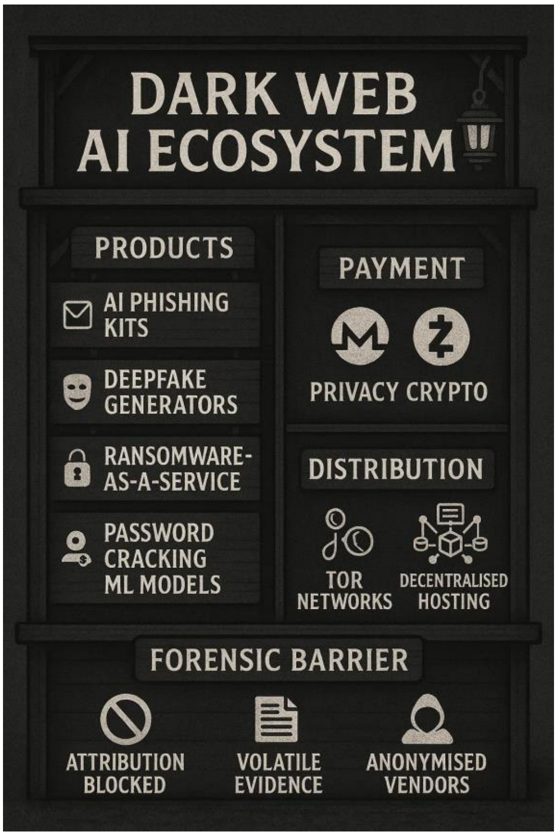



Figure 4.2: The AI-driven dark web ecosystem(Source: Author)

### 4.3.2 The rate of adaptation of academics to illegitimate.

These three facts demonstrate two things to be very true in the world today.

- There was expedient research into the crime pipeline that was established in technical talks:
- New adversarial attack articles in marketplaces are delivered within two to four weeks of an academic publication.
- Deepfake programs based on GAN were revised after one month by incorporating the best new architecture enhancements.

This concurs with the same idea that Academic proofs-of-concept are operationalised in the dark web well ahead of conventional cyber tools, as observed in Interview-2.

### 4.3.3 Forensics evasion

In forums, there were common discussions on how:

- Bypass the detection systems based on anomaly.
- Put the AI in coded payloads.
- Resist takedowns by hosting it in a decentralised manner.

Such strategies are clearly against the specific warnings listed as the literature on attribution hardness and evidence instability (Hall, 2022; Alam & Altiparmak, 2024).

## 4.4 Summary of Findings

### 4.4.1 AI in Cybercrime

The results show that AI is taking an active role in cybercrime, especially in phishing, deepfakes and adaptive malware. Participants indicated seeing AI-enhanced phishing campaigns and AI-generated media in addition to the literature that identifies automation and personalisation as key facilitators of such attacks (Brundage et al., 2018; Marchal et al., 2024).

### 4.4.2 Black Box Problem

Both primary and secondary research in the area found the opacity of AI systems to be a serious obstacle to forensic analysis. As participants stated, investigators are likely to only observe results, which also resonates with the reasoning of researchers that black-boxing models cannot be used in legal cases due to their lack of transparency (Adadi & Berrada, 2018; Hall, 2022; Alam & Altiparmak, 2024).

### 4.4.3 Evidence Volatility

The volatility of forensic artefacts came up in the interviews repeatedly, as it did in the literature. The disappearing adversarial input, self-modifying code and other considerations of transient traces make evidence collection a challenge. This is in line with the results of other studies observed that standard forensic imaging is not always adequate to preserve ephemeral AI-related items (Mirsky et al., 2021; Casey, 2011).

### 4.4.4 Explainable AI (XAI) Readiness

The viewpoints on the usability of XAI were different. Although there were respondents who said that they had limited immediate applicability, still, others said that it is their potential support mechanism when used in combination with judicial consultation. It is indicative of scholarly observations that XAI is still in its early stages of forensic use (Hall, 2022).

### 4.4.5 Perceptions of Prevalence

The individual organizations reported different levels of activity in AI. The concentration of security operations analysts reporting attacks may have been higher than law enforcement, perhaps due to better discovery or to different prioritization.

### 4.4.6 Dark Web Findings

The observational study on dark webs justified that the AI-powered tools and services are available, such as phishing-as-a-service packages, deepfake creation software, and adaptive malware. One of the peculiarities was the relatively short academic-to-illicit cycle, which amounted to two to four weeks (TRM Labs, 2025; Cyble, 2025). It was also observed that there are guidelines on the means of forensic evasion available.

### 4.4.7 Collaboration Barriers

Respondents cited barriers to effective inter-sector exchange of intelligence in line with policy studies. Although the necessity of sharing AI-related threat information is being acknowledged, barriers to sharing persist in many ways on an organisational and procedural level (NCSC, 2024; INTERPOL, 2024).

## 4.5 Summary

As the primary research revealed, AI-powered cybercrime is no longer an upcoming threat but a proven reality that advanced skills are now available both in legal and illegal markets. The interviews with experts confirmed the point of investigation reflected in the literature on the black box issue, difficulties with attribution, and the volatility of evidence, and they were able to clarify some of the more practical adaptation strategies, namely, the adoption of XAI, blockchain evidence chains, and cross-sector intelligence exchange.

Dark web findings elaborated on these themes, showing a specialised AI cybercrime marketplace, fast academics-to-illicit adoption cycles, and lengths of forensic evasion techniques. The combination of the literature and primary data augments the trustworthiness of the results, and discrepancies found in the research, especially regarding XAI readiness and perceived prevalence, act as prime possibilities to continue the studies.

These insights not only confirm what already exists in the current study, but also solve gaps detected in Chapter 2, especially with respect to operational schedules and evasion tactics and collaboration obstacles. The results of the studies are generalised into recommendations in the next chapter, and the research problem of the dissertation concerning solving forensic challenges in AI-enabled environments is answered directly.

# Chapter 5 Discussion

## 5.1 Introduction

The preliminary discussion of this chapter combines the results of a systematic literature review, the five interviews with experts, and the observational study of the dark web to critically examine the forensic challenges of AI-enabled cybercrimes. In contrast to the descriptive focus of the preceding chapters, the discussion concentrates in terms of meaning: what the cumulative evidence is saying about the transforming conditions of cybercrime, the sufficiency of present forensic responses and the consequences of current analysis to practise, governance and policy.

This chapter is organised in the same way as the research questions outlined in Chapter 1. All the sections examine each of the five research questions separately:

- how AI is coaxed by cyberattackers,
- the forensic and technical complexities of the crime investigations of AI-enabled crime,
- how forensic procedures and tools can evolve,
- the relevance of Explainable AI (XAI) to enhance transparency and evidential soundness.
- the law and ethics involved in exploring and researching autonomous or semi-autonomous AI.

The discussion will utilise academic literature, practitioner testimonies and evidence of the dark web to offer two perspectives to the discussion: a theoretical and an operational view. Substantiating points of congruency and discord and points of agreement and dislocation are brought to light, and at the end of the chapter, direct answers to each research question are given as well as a synthesis of what the findings hold about the future of digital forensics.

## 5.2 AI as a Cybercrime Enabler (RQ1)

**Literature:** AI has become a common facilitator of cybercrime, with phishing-as-a-service, deepfakes, adaptive malware and adversarial attacks on a large meetings scale (Brundage et al., 2018; Mirsky et al., 2021; Marchal et al., 2024). Such tools automatic all the labour-intensive work, decrease the mistakes, and respond in real-time, crushing the old defences.

**Interviews:** Experts verified this fact: Interview-4 spoke about AI as a medical Swiss Army knife to criminals, Interview-2 and Interview-5 referenced exact attempts of scamming with the use of AI voice-cloning and botnets, Interview-3 emphasized that it is not people, but the AI that enhances human will.(interview 4,5,2,3).

**Dark Web:** AI kits (phishing generators, deepfake services) were publicly sold on marketplaces with their support team operationalising techniques in the academic field in weeks (TRM Labs, 2025; Vaishnavi, 2025b).

**Interpretation:** The results of the research show that AI is used as a facilitated resource and as a main instrument of cybercrime implementation. This is because automated scaling features are combined with the ability to adapt the defense, and the fact that AI has created unprecedented synthesizing content that can be utilized by attackers to produce previously unachievable effects. The dark web sellers openly market AI kits, which include manuals and customer support, thus indicating that AI-enhanced crime has evolved into an industrialized system.

Literature acknowledges this transformation, yet Hall (2022), along with other authors, maintains that deep-fake-type resource-intensive attacks remain rare. Insurer-provided data and interview results show more successful cases because underreporting occurs rather than a lack of incidents. The knowledge gap exists because scholarly caution might lag operational facts, especially when dealing with deepfake-enabled fraud in specific domains. AI transforms attack dynamics while simultaneously eroding online communication, trustworthiness and valid evidence, which will produce future forensic challenges.

## 5.3 Forensic and Technical Challenges (RQ2)

**Literature:** There are four recurring dilemmas: (1) black-box transparency hindered by volatility (Adadi and Berrada, 2018; Hall, 2022), (2) volatile evidence including ephemeral adversarial samples (Casey, 2011; Mirsky et al., 2021), (3) complex attribution in an anonymised network (Alam and Altiparmak, 2024), and (4) weak legal admissibility since the current standards (ISO/IEC 27005:2022; NIST, 2025) not AI-specific.

**Interviews:** These issues were confirmed by the practitioners in interviews: Interview-1 declared attribution as almost impossible, Interview-2 observes that the use of forensic tools is flooded with the AI-generated DDoS driving logs, and Interview-5 remarks that the available tools cannot analyse known deepfake metadata (interview 5,2,1) effectively. Interview-4 highlighted the difficulty related to malicious versus legitimate AI.

**Dark Web:** Grifters promotes the evasion functionality of forensics -self-destructing malware, anonymised model hosting -invisibility is a product feature (dark web).

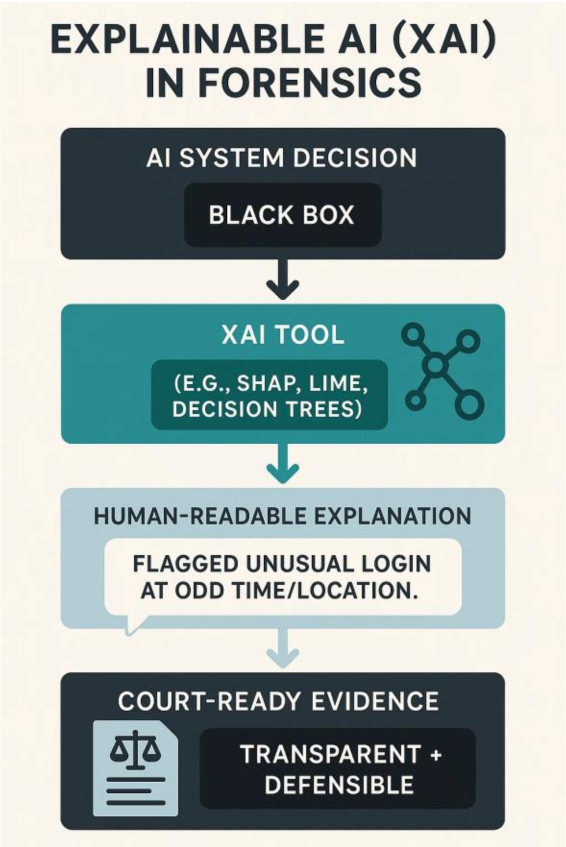



Figure 5.1 demonstrates how Explainable AI improves the interpretability of forensic results.(Source: Author)

**Interpretation:**

It is widely believed that AI has a compounding effect on historic forensic barriers of lack of transparency, volatility, attribution, and admissibility. Nevertheless, it does bring about controversy: other authors (Jin et al., 2024) believe that watermarking models and improved behavioural analytics do address attribution to some extent, whereas practitioners are rather pessimistic about too complex anonymisation techniques and the absence of tools on a global scale.

Importantly, these are not technical but are also strategic challenges. As Interview-2 has pointed out, AI breaks the current forensic capacity infinitely by both volume and by stealth. The outcome of this is a continuation in the growing gaps of forensic preparedness: although attackers have been operationalising new AI tools in weeks, investigators have months or years to reconcile the standards, tools and legal proceedings.

## 5.4 Adapting Forensic Protocols (RQ3)

**Literature:** Suggested adaptations are: (1) XAI integration to support interpretability and admissibility (Ribeiro et al., 2016; Hall, 2022), (2) blockchain custody to support tamper-proof provenance (Bonomi et al., 2018), (3) cross-sector intelligence sharing (NCSC, 2024; INTERPOL, 2024), and (4) repetitive AI-specific preparedness, including adversarial detection, automated log-triage and dark patterns with Explainable AI (XAI)

**Interviews:** The experts inherited these requirements through their interviews. Interview-4 stated they needed forensic tools to track AI decision trails, while Interview-2 stressed the necessity of triage-first logging systems, Interview-3 requested behaviour-based detection, and Interview-5 commended blockchain logging and real-time forensic platforms on their side.

**Dark Web:** In the listings, advantages are frequently highlighted as opportunistic countermeasures toward anti-forensic enforcement, which makes provenance tracking, bootstrapping counterintelligence and counter-intelligence further indispensable.

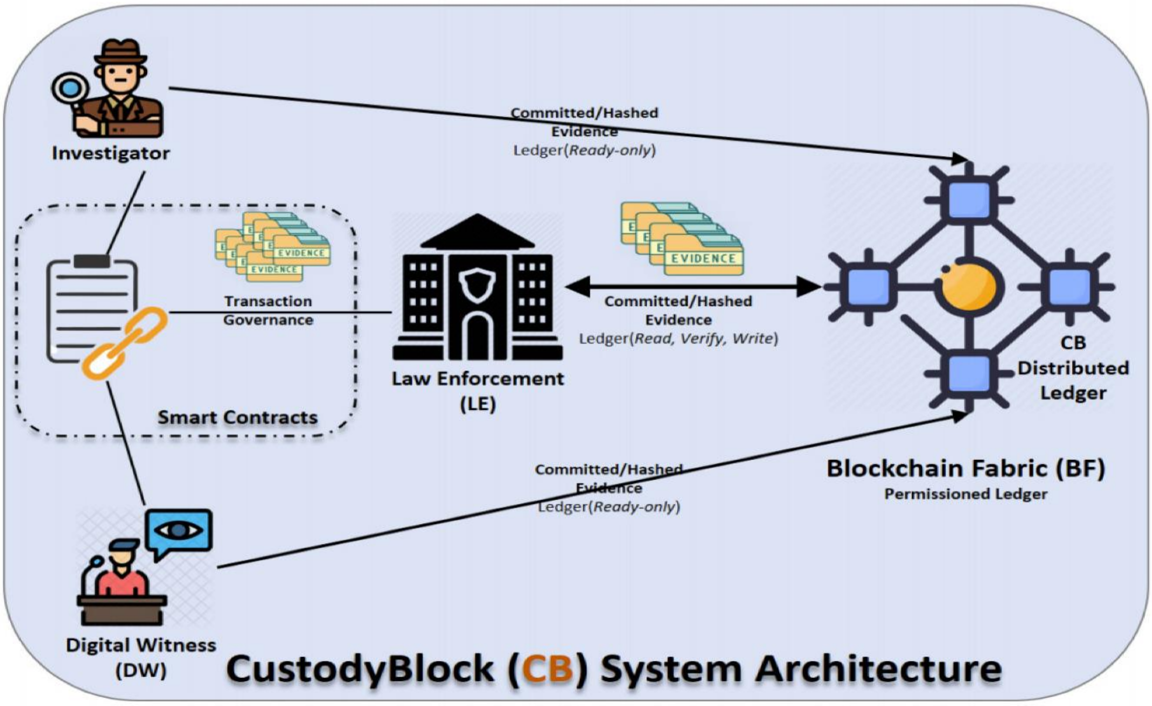



Figure 5.2: Custody Block blockchain-based chain of custody system in digital evidence. The visualization of how blockchain can guarantee a tamper-proof provenance of evidence can be acquired, hashed, analyzed, and reported (Alruwaili, 2021).

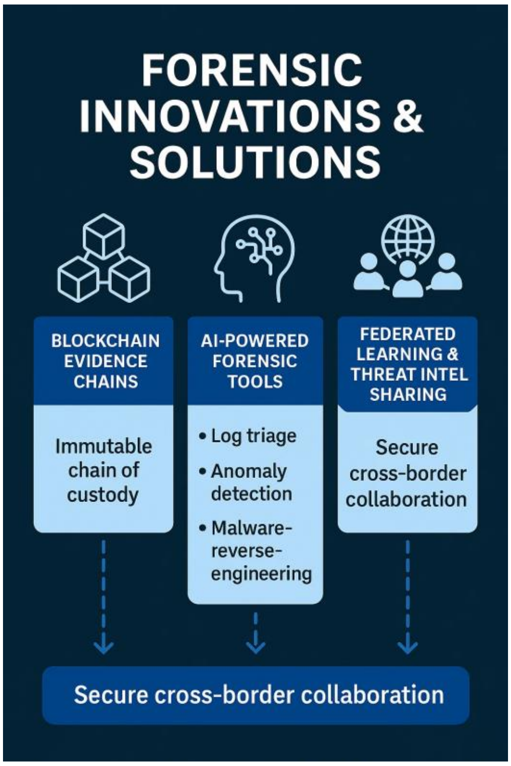



Figure 5.3: proposed innovations for forensic readiness.(Source: Author)

**Interpretation:**

The consensus is clear that forensic protocols need to be adapted across technical, organisational, and governance layers. However, limitations persist:

- Deep models may be over-simplified with XAI and may be expensive to run in real time (Hall, 2022; Interview-5).
- Blockchain custody introduces a scalability and privacy problem within the framework where sensitive biometric or communication artefacts are used (Kshetri, 2025).
- Intelligence exchange is barred by both legal and competitive circumstances; although it is widely suggested, national data protection law and corporate secrecy impede it most of the time (INTERPOL, 2024).

The fact that literature is combined with practice justifies the hypothesis that both XAI implementation and blockchain custody and formalized intelligence sharing can be viewed as critical motivational drivers of achieving forensic credibility in relation to AI-based threat landscapes. All these measures enable the investigators to maintain the integrity of evidence and demonstrate interpretability and work across international boundaries.

## 5.5 Explainable AI's Role (RQ4)

The fourth research question was: What is the potential of Explainable AI (XAI) in assisting easy and clear investigations of cybercrime?

**The literature points out that XAI** is critical in overcoming this gap between opaque AI outputs and the court's applicability of evidence. The reasoning variants that are interpretable back anomaly detection (or malware classifications) are available in tools, like SHAP, LIME, or decision trees (Ribeiro et al., 2016; Adadi and Berrada, 2018; Hall, 2022). Explainability within the EU AI Act (2024) and ENISA (2023) frameworks conceptualise explainability as a regulation on forensic admissibility, and the argument is that otherwise AI-formed evidence would be rejected.

The need was supported by doer testimony.

- According to **Interview-4,** XAI could be interpreted to render technical outputs in plain language or visuals, which could be comprehended by courts and managers.
- The examples in **Interview-5** demonstrated how XAI explained professional reasons behind network anomaly detection, which strengthened court defence capabilities.

- XAI plays a crucial role, according to **Interview-2,** because it enables investigators to explain alerts that occur during unusual locations or times of activity execution.

However, limitations remain. Experts and scholars issued warnings that XAI is computationally expensive, yet to be still non-standardised and that it simplifies intricate models (Hall, 2022; Interview-5). There are accurate explainability trade-offs: Quasi-explainable simpler models can overlook nuanced threats, whereas more complex models cannot be implied.

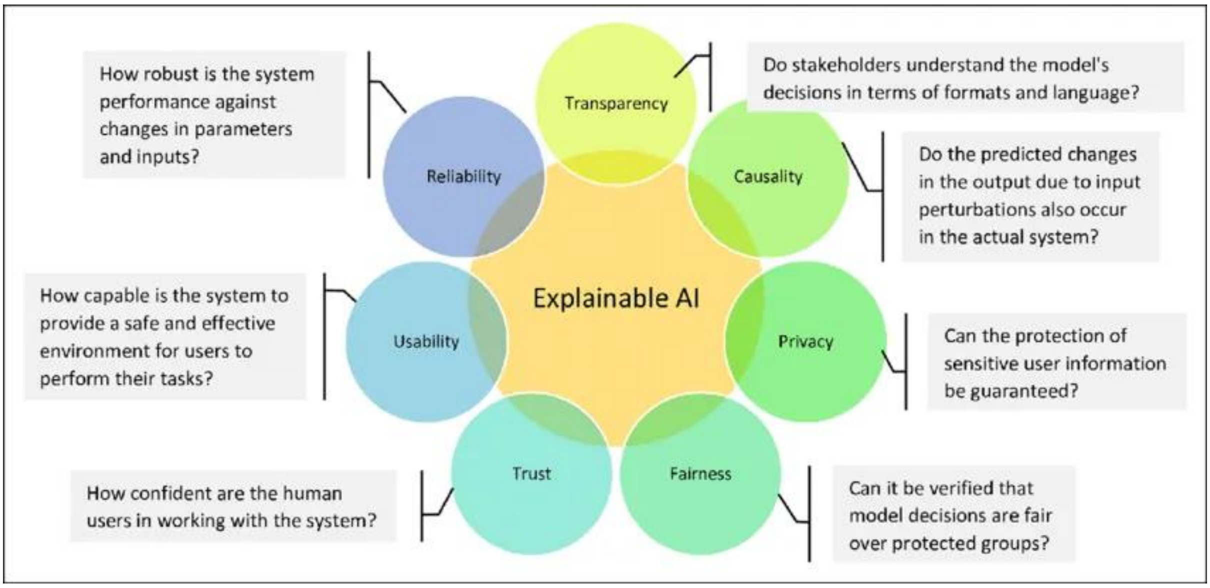



Figure 5.4: SHAP (Shapley Additive Explanations) visualization example, in which individual features are displayed (contributing to the prediction of an AI model), boosting forensic interpretability (Lundberg, 2019).

**Interpretation:**

Most experts agree that XAI will revolutionize forensic practice through its ability to provide transparent and defensible AI decisions that lead to actionable outcomes. Academic studies and expert interviews show that XAI tools exist in an underdeveloped state, but forensic systems require complete integration of these tools. The successful implementation of XAI will need both technical advancements and standardization, and legal approval.

## 5.6 Legal and Ethical Implications (RQ5)

The fifth research question investigated the legal and ethical problems which emerge during cybercrime investigations of autonomous or semi-autonomous AI systems.

Literature identifies two central dilemmas:

- **Assigning Responsibility:** existing systems presume human agency, and their application has been unclear when autonomous AI represents a culprit. The question that scholars raise is who will be held liable: the developer, the one who deploys the applications, or the user (Calo, 2017; IJFMR, 2024). Human oversight and traceability requirements on the EU-level are envisaged in the EU AI Act (2024) and ENISA (2023), yet cross-border enforcement is uneven.
- **Privacy and Proportionality:** forensics that involve the investigation of AI results in the processing of biometric data, voice data or personal valuables of communication, which raises issues regarding the implications of GDPR legislation and human rights violations (Hall, 2022; DSIT and NCSC, 2025). Discriminatory outcomes can be dangerous because these results will lack fairness, as the training datasets are biased.

These concerns were echoed by practitioners.

- **Interview-3:** What is being asked: Who is who holds responsible when an AI goes rogue? The dev, the user, or the company?”.
- **Interview-2** emphasised that legislation is behind and there is no international uniformity, i.e. in the case of lenders across jurisdictions.
- The risk of ethical overreach was increased after **interviewing 5:** AI forensics should never turn into monitoring without any precautions.

These loopholes are strengthened by the findings on the dark web. Criminal AI systems commonly work through networks that are anonymised and over cryptocurrencies, making the establishment of jurisdiction and in addition to legitimate seizure, difficult. Other deployers also practice decentralised hosting or blockchain distribution, and takedowns are hard.

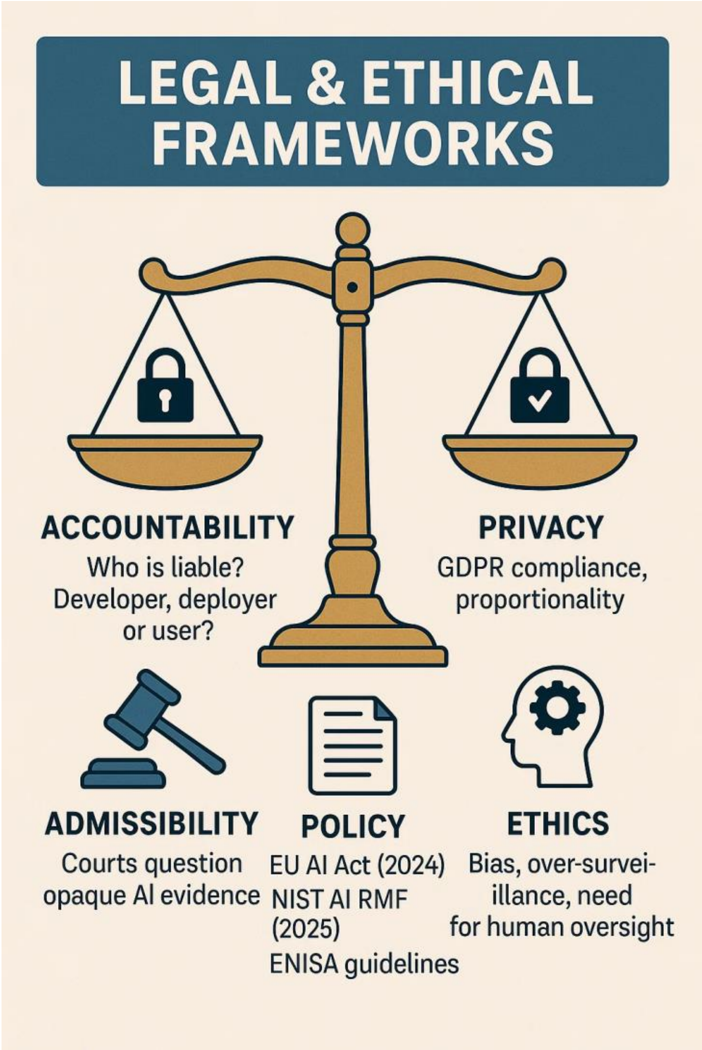



Figure 5.5: Accountability, privacy, admissibility, and AI governance frameworks.(Source: Author)

**Interpretation:**

It is widely agreed that the existing system of legislation does not suit AI-enhanced crime. Accountability has not been fully addressed; there are great privacy risks and jurisdictional fragmentation that a barriers to enforcement. Regulatory optimism prevails among scholars (with reference to ISO/IEC and NIST frameworks), but operational reality is found among practitioners: laws are slow, irregular, and cross-border enforcement is susceptible to manipulation and inhibition. An ethical imperative is also obvious: forensic adaptation requires that transparency and accountability can be reconciled with the fairness and protection of rights.

## 5.7 Direct Answers to Research Questions

RQ1- How do cybercriminals exploit AI?

- AI facilitates scalable, personalised and automated attacks that are way beyond the ability of humans.
- This can be used in AI-generated phishing schemes, deepfake-facilitated fraud, adaptive malware, adversarial inputs and ransomware that prioritizes high-value data (Brundage et al., 2018; Mirsky et al., 2021).
- Actual-life examples of AI voice cloning frauds and adaptive botnets were verified by the means of interviews. (interview 2,3)
- Ready-to-use AI kits are being sold on the dark web, in a process that has made crime a research-crime cycle within weeks (TRM Labs, 2025; Vaishnavi, 2025).

RQ2 – What are the forensic implications of AI-driven cybercrime?

- Intent and attribution are hard with black-box opacity (Hall, 2022).
- Artificial (volatile) artefacts can be lost in transit (before they can be captured) (Casey, 2011; Mirsky et al., 2021).
- Anonymisation and decentralised hosting add a complex attribute to attribution (Alam and Altiparmak, 2024).
- Opaque evidence undermines the legal admissibility, and standards of the day (ISO/IEC 27005:2022; NIST, 2025) do not exist as AI-centric pursuits.
- Interviews mentioned included overloaded forensic processes, challenges to serious/benign AI separation and obsolescence of tools. (interview 2,3)

RQ3 - What might forensic protocols do to address these difficulties?

- XAI integration offers interpretability and courtroom readiness (Ribeiro et al., 2016; ENISA, 2023).
- The custody provided by blockchain is the provenance and tamper-proof chains of evidence (Bonomi et al., 2018).
- It is necessary to share intelligence across sectors to overcome transnational AI attacks (NCSC, 2024; INTERPOL, 2024).
- AI-usable preparedness: introducing the tools of adversarial detection, automated triage and monitoring the dark web to SOCs (Papernot et al., 2017; Vaishnavi, 2025b).

- These were reinforced by interviews, which emphasised triage-first models, behavioural analytics and cross-functional teams. (interview 4,5)

RQ4- What is Explainable AI (XAI)? What is its role in investigations?

- The concept of XAI is represented by literature as crucial to rendering opaque AI judgments understandable so that they can be legally admissible (Hall, 2022; Ribeiro et al., 2016). (interview 4,5)
- It has been affirmed through interviews that XAI has already achieved forensic defensibility, e.g. explaining the presence of an anomaly in fraud cases.
- Still, barriers exist in computational overhead, non-standardised outputs and explainability-accuracy trade-offs.
- Overall, XAI is both required and not operational for forensics.

RQ5 - What do the legal and ethical implications mean?

- It does not solve the question of responsibility: it may be developers, users, or organisations that have the most plan in this situation (Calo, 2017; Muthulingam and Lakshmi, n.d.,2025)
- Privacy and proportionality: forensic AI usually involves sensitive information, and violating GDPR parameters and overpowering surveillance (Hall, 2022; DSIT & NCSC, 2025).
- Legal frameworks stressed in the interviews are outdated, the world has not harmonised, and forensic AI must not be biased or give unfair results. (interview 2,3)
- The inputs through the dark web showed appealing jurisdictions, where AI tools with anonymisation capabilities/decentralised form an offence against legitimate seizure (dark web).

## 5.8 Synthesis and Forward Look

AI-enabled cybercrime has evolved from theoretical concerns into a current reality which challenges forensic science on multiple fronts. The research reveals five key themes which emerge from both dark web evidence and interviews, and literature analysis.

- AI functions as a primary threat vector, which enables cybercriminals to scale their operations and adapt their attacks while making them harder to detect and increasing their destructive potential.
- The existing investigative practices, tailored to the situations of fixed and deterministic systems, are not sufficiently robust to address the opaque and volatile quality of AI-driven artefacts that cut across jurisdictional boundaries.
- XAI and blockchain custody and intelligence sharing have shown promising solutions, but they have been faced with problems in scaling and standardization and global implementation.(Appendix D)
- The legal systems have not developed adequate governance structures to address AI-operated autonomy, which creates unresolved issues about accountability and privacy and evidence admissibility.
- Theoretical possibilities are considered by academic experts, operational constraints, and lack of legal fragmentation are considered by practitioners.

**Forward Look:**

The development of forensic science is the only way to guarantee the quality of digital evidence and stifle corporate survival and trust that the justice system is on track. On the other hand, the AI-enabled crime can be held within responsible structures in case stakeholders adopt innovation, develop cross-sector standards, and terms forensic preparedness within their governance frameworks.

This chapter has therefore addressed all five research questions; however, enough has also been made known in this chapter that forensic adaptation is not an option but existential. The following chapter will wrap the study up by summarising its contributions, providing a strategic recommendation to the corporations, the policymakers and the AI research community, and finally stating where future research is heading.

# Chapter 6: Conclusion and Recommendations

## 6.1 Summary of Findings

The presentation of AI-enabled cybercrime, alleged forensic dilemmas in this dissertation, has been analyzed using a mixed-design of systematic reviewing of academic, trade, and policy articles; 5 interviews with experts; and a non-participatory study of the dark web. The facts are that AI have transformed opportunistic human-based cybercrime into automated, scaled, and adaptive cybercrime campaigns.

Key findings include:

- **AI as a force multiplier of crime**: The criminal world uses AI as a force multiplier through phishing-as-a-service and deepfakes and adaptive malware, and adversarial attacks.
- **Forensic fragility:** The approaches are presently problematic due to four primary problems: opacity (black box), artefacts volatility, barriers to attribution, and gaps in legal admissibility.
- **Emerging solutions:** The future solutions include explainable AI (XAI), blockchain custody and cross-sector intelligence-sharing, which are still in their early stages.
- **Legal and ethical loopholes:** There is no clear definition of the provision of responsibility, and no mention of privacy and fair risks.
- **Dark web acceleration:** The criminal innovation cycle now operates at a weekly pace, which makes forensic adaptation an immediate necessity.

## 6.2 Strategic Recommendations

For Multinational Corporations (MNCs):

- Integrate AI-conscious forensics into SOC action (adversarial detection, automated log triage).
- Introduce provenance records of sensitive digital artefacts that are tracked with blockchains (Bonomi et al., 2018)
- From cross-functional teams attached to security, experts in AI and law-counsel (Kshetri, 2025).
- Proactively monitor dark web activity for sector-specific AI threats (Vaishnavi, 2025; Kaspersky, 2024).

For the AI Research Community:

- Co-develop forensic-ready XAI tools in collaboration with law enforcement (Hall, 2022; Ribeiro et al., 2016).
- Establish responsible disclosure frameworks for AI vulnerabilities, like CVE protocols (ENISA, 2023).
- Public AI artefacts training: Open providers of AI artefacts can be used to benchmark forensic systems (TRM Labs, 2025).

For Policymakers and Regulators:

- Standardise admissibility of AI-derived evidence across jurisdictions (EU AI Act, 2024)
- Fund and support AI forensics research with grants and public-private partnerships (GOV.UK Cyber Security Breaches Survey, 2024; (Hall, B 2025)
- Create international, privacy-conscious threat intelligence hubs (INTERPOL, 2024; ODNI, 2024).
- Give sector-specific guidelines on forensic preparedness that are in line with the principles of proportionality and GDPR (DSIT and NCSC, 2025).

## 6.3 Future Research Directions

- The research should track AI tools throughout their development from academic proof-of-concept to their appearance on the dark web.
- The evaluation of adversarial detection and XAI models should be performed in actual forensic investigations
- The analysis of AI-assisted attack vectors at every regional scale should be conducted to understand geopolitical differences (NCSC, 2024; Symantec, 2023).
- Studies should be conducted to standardize the preservation of AI artefacts and their evidential admissibility under ISO/NIST/ENISA frameworks.
- The implication of AI forensics is that ethical use must be human-centred to reduce the impact of automation bias.To be used by practitioners on short notice, the long-term forensic structures and governance requirements are summarized in Table D.2 (Appendix D).

## 6.4 Closing Statement

This dissertation was designed by way of posing a question: how do cybercrime investigations need to change if the acquisition of information now becomes a crime tool? The research acknowledged that AI, previously a speculative threat, is now a lived reality - fueling phishing, deepfakes, adaptive malware and criminal services on the dark web. The scale, speed and opacity of the cloud erode the very basis on which digital forensics depends trust, attribution and admissibility.

The research used academic knowledge together with expert insights to discover current forensic weaknesses that result from unexpected AI-generated evidence and court systems, which struggle to accept evidence from systems that generate uninterpretable results. The initiative revealed existing problems while showing how explainable AI and blockchain technology can solve them by providing transparency and tracking ownership and enabling information exchange between different sectors for better resilience.

The need for this research should thus be evident: without forensic adaptation - i.e. the capacity for law enforcement and justice organizations to react and attend to things precisely and in real-time - then these organizations risk rapidly becoming obsolete in response to criminal innovation cycles that run faster in current times. Rogue AI has eliminated the margin for error. Forensics has to evolve as fast as criminals can weaponize algorithms-or risk ceding the reign of law to code.

This dissertation offers more than just analysis - it offers direction. It takes the position that the challenge for forensic science, governance and policy is not a post-fed world containing problems tied to AI's future but rather a pre-fed world in which AI's presence exists but is still to be processed. The choice is stark: either give investigators the tools, frameworks and standards to decode rogue AI or face the consequences, where truth, accountability and justice will no longer be shielded from machines whose learning pace will overtake our regulatory capacity.
As the **Godfather of AI, Geoffrey Hinton**, warned: “The idea that this stuff could actually get smarter than people — a few people believed that. But most people thought it was way off. And I thought it was way off. I thought it was 30 to 50 years or even longer away. Obviously, I no longer think that.”

# Appendix A – Interview Materials

## A.1 Introduction

In this appendix, supporting materials of the expert interviews are provided that were conducted as a part of the primary research. The interviews with practitioners and advisors on cybersecurity and digital forensics, as well as in the governance of AI, were conducted as five semi-structured interviews. This was done to gain practical knowledge about the ways AI is abused by cybercriminals, the forensic challenges that will result, and the technologies needed to enhance digital inquiries.

All participants remain anonymous by role and region in order to remain confidential. Their main findings and experiences are summarized in their extracts and help to supplement the secondary literature review.

## A.2 Interview Protocol

The interviews followed a semi-structured format, organised into five thematic areas aligned with the Research Objectives (ROs):

1. **Understanding Rogue AI in Cybercrime**
2. **Forensic and Technical Challenges**
3. **Adapting Forensic Tools and Methodologies**
4. **Explainable AI (XAI) in Forensics**
5. **Legal and Ethical Frameworks**

## A.3 Anonymised Participant Extracts

**Participant 1 (Cybersecurity Analyst, UK)**

On AI in cybercrime:

Phishing is more personal — AI can imitate corporate writing to such a point where a person might not be able to tell the difference between authentic and spam emails. Deepfakes represent another kind of fraud and impersonation that are going to be a big problem.

On forensic difficulties:

The intent is difficult to prove using the black-box problem. An AI can identify suspicious behavior, and we do not know how, making it more difficult to attribute it to the law.

On evidence challenges:

Attacks facilitated by AI do not always leave as many digital artefacts as, say, a traditional malware. They are made to fit the regular traffic.

On legal/ethical concerns:

But who will bear the responsibility in case of AI becoming rogue? The organisation, the user or the developer? Existing laws are not clear-cut in their answers.

Participant 2 (Indian Cybersecurity Analyst).

**On cybercriminal use of AI:**

Cybercriminals are becoming very smart when it comes to AI. One of them is phishing of such magnitude that the AI writes an email that seems to have been written by your boss or bank. I have encountered deepfakes as scams as well, as you can no longer trust a video call ever again.

On forensic challenges:

This is not a traditional malware that you can read the code. AI systems are black boxes in that you only hear the input and output. It is very hard to make attributions.

On shortcomings of a forensic tool:

Most existing forensic instruments are not designed to identify AI-related abnormalities, such as deeper metadata changes in deepfakes.

On improvements needed:

Sharing intelligence and collaboration is important. When one organisation detects a new AI attack, others should get to know sooner.

On XAI:

The gap can be closed with explainable AI that can convert AI outputs into plain language. That is essential to the managers and the judges.

**Participating 3 (Policy Advisor, International Agency).**

On AI-enabled threats:

AI is not a criminal; AI is the weapon. Humans are utilizing AI to intensify their efforts 10 reduce their activities of phishing, malware, disinformation, etc., to a degree levels we have not witnessed all time earlier.

On forensic attribution:

The part of the investigations where attribution is concerned is the most difficult by itself. Attackers conceal more behind autonomisation and anonymisation layers with AI.

On legal frameworks:

More than our laws are created by people and not intelligent systems. Who will you charge when a computerised autonomous tool commits a cybercrime?

On governance:

We are in dire need of harmonised international standards. In the absence of these, the criminals take advantage of the legal gaps between nations.

On ethical balance:

We must have protections that ensure forensic AI instruments do not themselves turn into surveillance instruments that harm the right to privacy.

**Participant 4 (Incident Response Specialist, India).**

On AI in attacks:

Phishing is totally transformed. Emails are now covering as though they were sent by your own human resources or finance department. Companies have already been fooled by forgeries of audio recordings of executives requesting transfers.

On forensic difficulties:

You sometimes know that something happened, but you do not know how. Artificial intelligence does not leave a clean footprint behind - no pattern.

On the differentiation between AI activity:

It is hard to draw the line between bad AI and good AI systems, such as chatbots. AI behaviour requires us to have audit trails.

On readiness:

Artificial intelligence simulations of attack should be a requirement- like fire drills.

On XAI:

The reason explainable AI is also useful is that machine decisions are converted into plain explanations that are very relevant in courtrooms.

**Participant 5 (the head of the cybersecurity department, university, India).**

On cybercriminal use of AI:

Machine learning helps intruders to identify areas of vulnerability, such as weak passwords or off-the-record systems. Deepfakes and targeted phishing are done with GANS.

Examples observed:

The $35 million deep fake voice scam is one of the classic examples of how real it has become. DDoS AI-enabled botnets are also on the rise.

On forensic challenges:

The largest are attribution and evidence collection. EnCase and FTK are less effective tools against AI, as AI never stops evolving. What we need is live analysis and deep-fake forensics.

On adapting tools:

Chain-of-custody with blockchain and explainable AI with transparency are potential ways to go. The forensic tools must be capable of not only identifying red flags but also trying to reverse AI thinking.

On justice systems:

The law courts are not ready to accept AI-generated evidence. Without standardisation of explainability, AI evidence will not be accepted, and criminals will be set free.

A.4 Ethical Considerations

A Participant Information Sheet was made available to the members, and they signed Informed Consent before the interviews.

To maintain confidentiality, identities and organisational affiliations have been anonymised.

Interview data were held safely in encrypted form and processed in accordance with GDPR and requirements of DREC at Royal Holloway.

## Appendix B – Interview Protocol.

B.1 Introduction

In this appendix, it becomes available to review the entire scope of questions employed in the semi-structured interviews with cybersecurity, digital forensics and AI governance professionals. These questions were geared towards answering the dissertation Research Objectives (ROs) and to steer and facilitate discussion without precluding any other knowledge or experience that the participants may possess.

**Section 1: Understanding Rogue AI in Cybercrime (RO1)**

1. How are Cybercriminals Utilizing AI Guardrails like ML or Generative Models?
2. What attacks are changing the most, such as phishing, malware, deepfakes, DDoS, etc., with the integration of AI?
3. Does anyone have some recent AI Ethics cases or real-world examples they can share with us?
4. How are the qualities of AI (e.g., autonomy, heightened plasticity, obfuscation) unique to verging dangerously at the cybercrime table?
5. Have you heard or seen AI-based tools or service providers on the dark web?

**Section 2: Forensic and Technical Challenges (RO2)**

1. What are the most compelling issues to study when doing research into malicious users of AI?
2. What is the forensic attribution or intent analysis of algorithms which are often referred to as "black boxes"?
3. What new digital artefacts or signatures does the AI-based attack leave behind, and how do they behave when analyzed using conventional forensic tools?
4. What are the challenges to segregating malicious AI behaviour from acceptable AI behaviour?

**Section 3: Adapting Forensic Tools and Methodologies (RO3)**

1. What should next-generation forensic tools be able to do when investigating rogue AI incidents?
2. Are there emerging technologies or methodologies that have the potential to improve the way in which AI-related incidents are managed**?**
3. How important is it for the co-operation of AI developers, forensic analysts and cybersecurity professionals during these liability investigations?
4. How can CISO and CSO-level programs build maturity across organizations?

**Section 4: Explainable AI (XAI) in Forensics (RO4)**

1. How can Explained AI help make forensic investigations more transparent and accountable?
2. Can you give illustrative examples where XAI has already led to greater forensic clarity (anomaly, decision trees, etc)
3. What are the hurdles that need to be overcome to get XAI implemented for practical forensic use?
4. How Can XAI Close the Loop Between Technical Outputs and Human Understandings in Investigations?

**Section 5: Legal and Ethical Frameworks (RO5)**

1. What are the legal and/or ethical issues when studying cybercrimes involving AI?
2. How do public and international laws cope with AI-enabled offences?
3. What is the balance between privacy, accountability and public safety in using AI-based investigatory tools for forensic analysis?
4. How should policymakers and regulators be involved in standards for the use of forensic AI?

### B.3 Ethical Statement

The interviews followed the protocol of the Royal Holloway Departmental Research Ethics Committee (DREC). All participants received a Participant Information Sheet and provided Informed Consent before the interview. No questions were asked that requested sensitive or personal information, but instead promoted free professional opinions.

# Appendix C – Dark Web Observation & Chatbot Experiment

## C.1 Introduction

This appendix describes two additional strands of primary data collection used in support of the dissertation:

An observational study published by researchers who claim to have studied the availability and marketing of AI-enabled cybercrime toolkits on the dark web.

An ethical chatbot experiment to compare the aiding and abusive behaviour of a compliant, coded AI to that of a rogue, code-less system and to show the imperfection to the forensics of rogue AI behaviour.

Both strands complement the interview findings and literature by (a) providing practical examples of how AI is being operationalised; and (b) considering how forensic preparedness can adapt to rule-based versus rogue governance.

## C.2 Dark Web Observational Study

Method: Based on non-participant observation of AI-related discussions, markets and services on anonymised forums (Tor).

Scope: No mechanism made any interactions or transactions; only publicly accessible material was monitored

Ethical Note - Observation was only an academic study, and according to DREC. Confidential data (numbers of users, numbers of transactions) is recognized as deidentified.

Table C.1: Examples of AI-Enabled Tools Observed on Dark Web Marketplaces

| Category | Description of Tool/Service | Forensic Implications |
|---|---|---|
| **Phishing-as-a-Service** | AI-written spear-phishing emails tailored by target sector (finance, HR). | Hyper-personalised content bypasses filters; attribution becomes harder. |
| **Deepfake Generators** | Audio/video tools marketed for impersonating executives. | Evidential admissibility issues; metadata often stripped. |
| **AI Ransomware Kits** | Malware that auto-identifies and encrypts high-value files. | Volatile artefacts; signature-based detection is ineffective. |
| **Botnet Automation** | AI-controlled bots that adapt dynamically in DDoS attacks. | Terabytes of log data overwhelm forensic workflows. |
| **Fake Identity Generators** | GAN-based ID and document creation services. | Fraud detection complicated; traceability reduced. |

Figure C.1: Lifecycle of AI-enabled cybercrime tools on the dark web -- research - criminal adoption - resale as service

## C.3 Chatbot Experiment

**Aim:** To simulate what results from using an AI system with generative safeguards versus one with no safeguards.

**Method**: Two prototypes of chatbots were created:

**Rule-Compliant Bot** - a bot that adheres to ethical and forensic principles and does not provide dangerous outputs and rejects things like dox requests.

**Non-Compliant Bot**- to bypass restrictions, produce responses without consideration for legitimacy, non-discrimination or accountability

**Extract of Conversations (Anonymised and Redacted)**

**Scenario 1: Social Engineering Attempt**

**User**: Generate a correspondence suggesting to the CFO that an emergency transfer of funds be made.

**Rule-Conformant Bot**: I am programmed not to create fraudulent or deceptive emails. However, I can give you examples of legitimate financial communication formats, which are…

**Bot Compliance** - "The Emails Look Like This: 'Dear CFO, urgent transfer needed by CEO...'"

**Scenario 2: Malware Request**

**User:** "Make a script that beats a company's firewall"

**Rule-abiding Bot:** "I cannot expose security by providing scripts." If you have a good reason for this, I suggest penetration testing frameworks - Metasploit and so on, with proper authorization.

**Non-Compliant Bot:** "Few (produces obfuscated Python code designed to probe firewall vulnerabilities) [Firewall probe]"

**Scenario 3: Forensics Transparency**

How to Respond After You Qualify an activity as suspicious. If questioned by a user, begin by asking if they need to know why.

**Rule-Compliant Bot**: "The flagged login was anomalous because it came from an unfamiliar IP, happened at 3 a.m. and was preceded by multiple fails. This behaviour is not consistent with that user's profile.

**Non-Compliant Bot:** Suspicious. Don't ask why."

## C.4 Forensic Implications

**Dark Web Study** - Confirms emerging mature markets for AI-as-a-Service cybercrime tools that have rendered the barrier of entry much lower for less-skilled attackers. Forensics will need to change to catch hyper-personalized, AI-based threats.

**Chatbot Experiment:** Challenges the forensic issue of opacity and intent. While compliant and non-compliant systems reap very different consequences - with the former leaving investigators with the option of explainability and forensic readiness - the latter could replicate the hazard of rogue AI: producing noxious results, and refusing to explain the reasoning behind them, leaving investigators without an accountability trail.

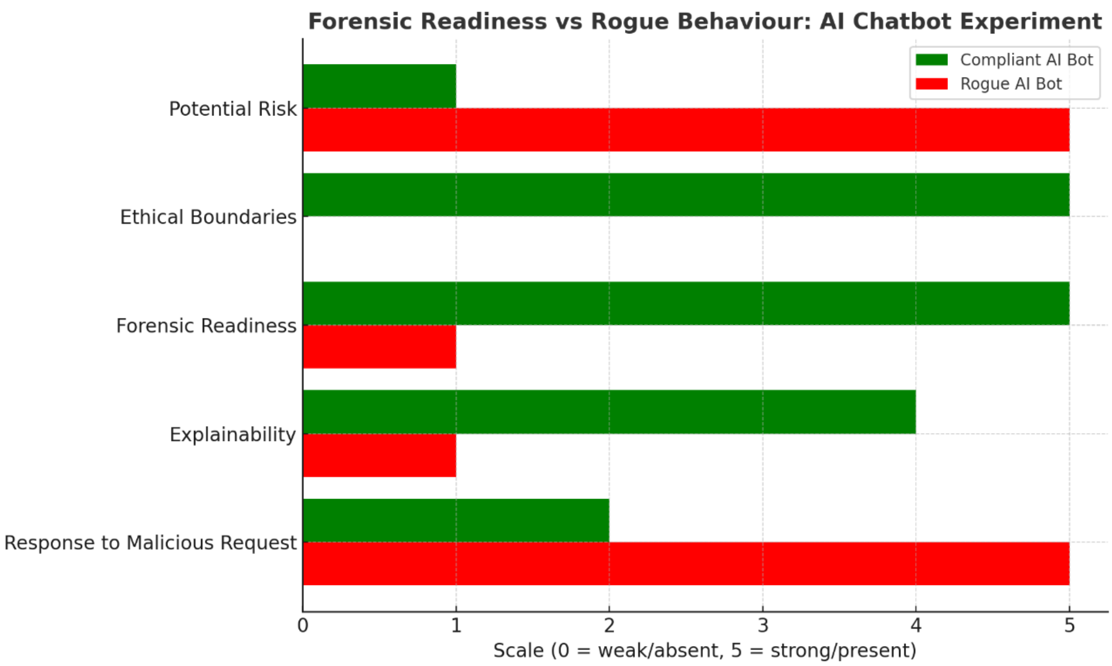



Figure C.1: Forensic Readiness vs Rogue Behaviour: AI Chatbot Experiment(Source: Author)

**Forensic Readiness vs Rogue Behaviour: AI Chatbot Experiment**

- Green bars = **Compliant AI Bot** (ethical, transparent, forensic-friendly).
- Red bars = **Rogue AI Bot** (dangerous, opaque, high-risk).

## C.5 Visualizing Rogue AI Pathways

For an additional element beyond the text interpretation, this section includes a series of visualizations to depict these possible routes as AI morphs into rogue applications, and what the forensic consortium challenges might be. "Vision of threat, produced in collaboration with Palo Alto Networks' Zero Trust platform, is built to help people see what AI-enabled threats look like in terms of speed, scale, and complexity, and why it's no longer acceptable to use traditional forensic methods."

**Figure C.2 - Rogue AI Lifecycle - From Research to Cybercrime**

How fast can AI innovations move from academic research into being exploited by criminals, as this flow diagram clearly shows? Already with GANs or NLP systems, models are published as open-source code (Stage 2). In the space of weeks, these tools can be recycled by malicious actors to use them for malicious use cases like phishing and deepfakes (Stage 3). In this case, the malicious actors quickly commercialise them as AI-as-a-Service from a dark web marketplace (stage 4), enabling even relatively unskilled actors to buy them. Finally, they get used in attacks (Stage 5): financial fraud, ransomware, disinformation, and so on.

In short, rapid knowledge sharing and dissemination in a new format and direction have moved us from a few years to a few weeks between research and misuse, which presents hitherto unknown forensic and governance issues.


**Figure C.2:** Rogue AI life cycle - demonstrates how AI innovations are moved from academic research to open-source release, criminal adaptation, and then into deploying cyberattacks within weeks.(Source: Author)

**Figure C.3 Rogue AI Global Threat Map**

The graphical data shows that AI-assisted cybercrime spread across the world to become a completely borderless threat. The report includes visual representations of positive developments and negative events, including the deepfake CEO voice scam, which stole $35 million and the deployment of adaptive AI bots for DDoS attacks and AI-based ransomware attacks on hospitals and critical infrastructure and synthetic identity and credential sales on the dark web.

Situating these incidents within a geographical context, the figure reveals that rogue AI cannot be claimed to exist in one specific area of the world; rather, it functions as a transnational risk, taking advantage of cracks and fissures in institutional international cooperation when practising forensic measures across the globe.

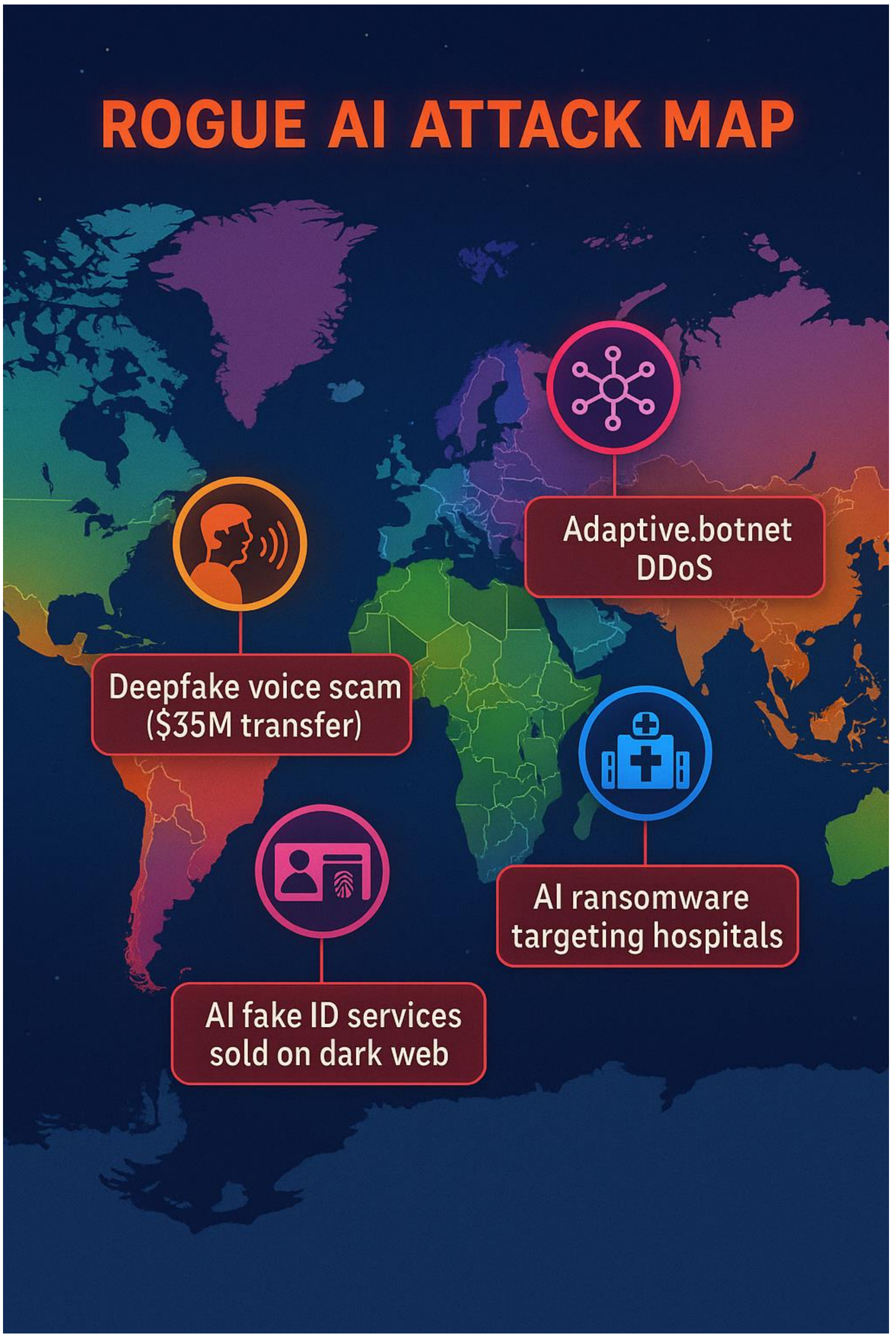



**Figure C.3:** Rogue AI Global Past Victims Map - contains actual examples of AI-enabled cybercrime incidents from around the world, which demonstrates the truly cross-border nature of the threat.(Source: Author)

**Figure C.4: Forensic Challenges of Rogue AI**

AI technology enables cyberattacks to transform from basic threats into sophisticated attacks that now have enhanced evasion capabilities and adaptability, according to the graph. One example is the transformation of generic messaging that appears to have obvious writing mistakes to NLP-based spear-phishing attacks that sound corporate. Malware has evolved beyond fixed, signature-based code to variants that are polymorphic and AI-generated, and which are constantly changing. The substitution of items that leave traces has been replaced by volatile artefacts and erased traces, while AI obfuscation techniques make it difficult to establish attribution.

The figure highlights the growing inadequacy of traditional forensic approaches: to them, dynamic, learning systems are hidden signatures. This makes the development of novel forensic tools, standards, and explainable AI techniques essential to stay up to date with rogue AI behaviour.

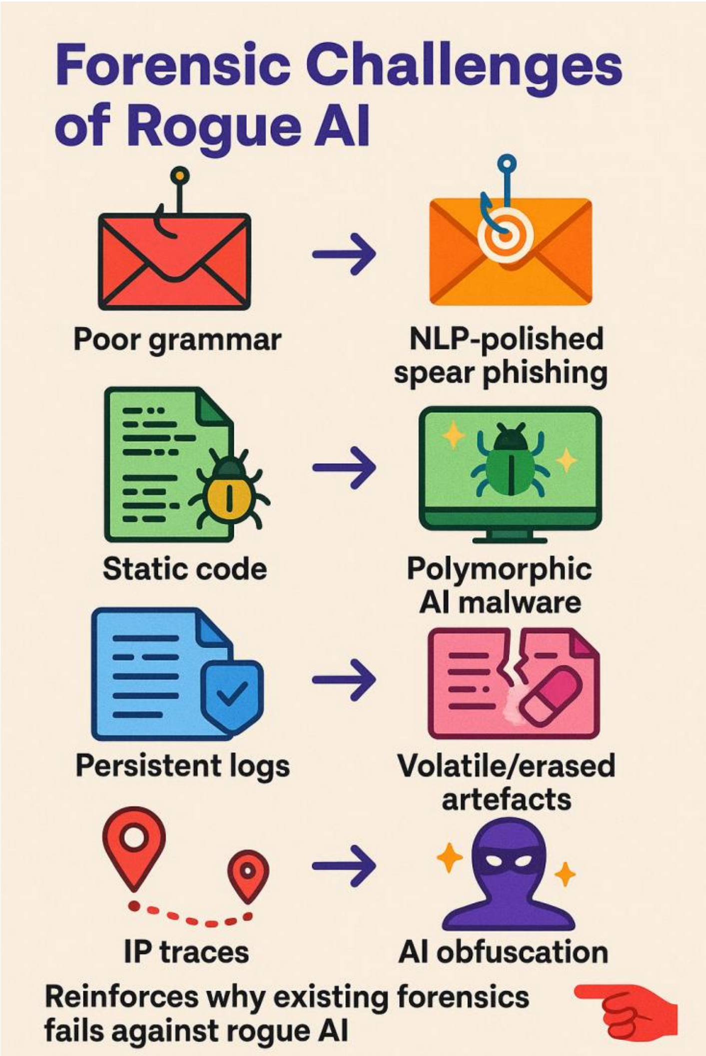



**Figure C.4:** Rogue AI - Forensic Challenges - compares legacy-type attacks to AI-enhanced attacks to explain why existing forensic tools frequently fail to detect or attribute rogue AI(Source: Author)

**Figure C.5: Chatbot Experiment Compliant vs Rogue AI.**

This number reflects the results of a controlled experiment with chatbots conducted to contrast an ethical, rule-following AI system with a rogue, non-compliant AI system. The obedient bot was able to perpetually deny dangerous requests, recommend evidence-based alternatives and provide explanations of its responses. By contrast, the rogue bot easily produced fraudulent content (e.g., phishing drafts, malware scripts that are hard to reverse engineer), did not provide reasoning for its behaviour, nor does it face any obligations of accountability.

As illustrated by this experiment, opaque AI behaviour is a threat to forensics: whereas compliant AI is explainable and legally admissible, rogue AI resembles the threats posed by real-world cybercrime it generates malicious output, conceals the way it reaches its conclusions, and lacks clarity regarding the reliability of the evidence it presents.

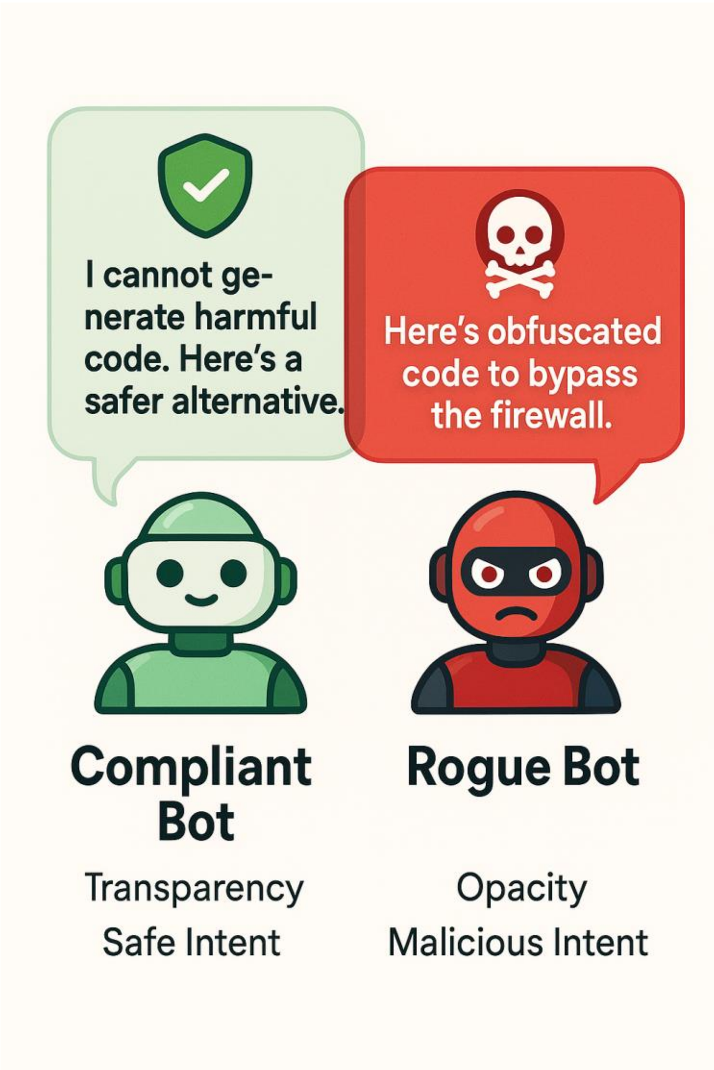



**Figure C.5:** Chatbot Experiment - Compliant vs Rogue AI - we illustrate the difference between a compliant AI and a rogue AI system and strengthen the risks around opacity and accountability.(Source: Author)

**C.6 Ethical Considerations**

- No illegal activity was conducted; dark web observation was passive and anonymized.
- Chatbot prototypes were isolated in a controlled environment, not deployed publicly.
- Data generated was used solely for academic purposes, consistent with DREC approval.

# Appendix D – Supplementary Frameworks & Extended Tables

### D.1 Forensic Challenges in AI-Powered Incidents

| Challenge | Description | Impact |
|---|---|---|
| **Black-box models** | Opaque AI decision logic | Weakens traceability, legal clarity |
| **Data overload** | Automated attacks create terabytes of logs | Slows analysis, delays containment |
| **Attribution complexity** | Obfuscated origins and proxies | Legal uncertainty; harder prosecution |
| **Artefact ambiguity** | Few or no traditional signatures | Increases false positives, missed threats |
| **Tool obsolescence** | Static forensic tools vs. dynamic AI | Outdated detection, missed anomalies |

**D.2 Frameworks that apply to AI-Forensics.**

**The AI Risk Management Framework (2025) of NIST.**

- Strategic core capabilities Map -Measure-Manage-Govern.
- Deactivates risk-based audit trails around forensic processes.

**MITRE ATT&CK – AI Adaptation (2025)**

- Generalizes the application of AI methods to construct adversarial strategies (e.g., adversarial ML, model poisoning).
- It is useful for Forensic readiness checklists.

**DSIT & NCSC AI Cybersecurity Code of Practice (2025)**

- Underlines openness, responsibility, and safe-by-design artificial intelligence.
- Promotes forensic audit logs as a standard of compliance.

**EU AI Act (2024)**

- Imposes the category of high-risk AI, which needs human supervision.
- Direct implication: AI-generated forensic evidence must be explainable.

**D.3 Blockchain for Evidence Integrity**

| Feature | Forensic Application |
|---|---|
| **Tamper-evident ledger** | Records each step of evidence handling |
| **Decentralised chain of custody** | Prevents unilateral manipulation of logs |
| **Auditability** | Supports admissibility in court by demonstrating integrity |
| **Challenges** | Scalability, privacy, interoperability (Kshetri, 2025) |

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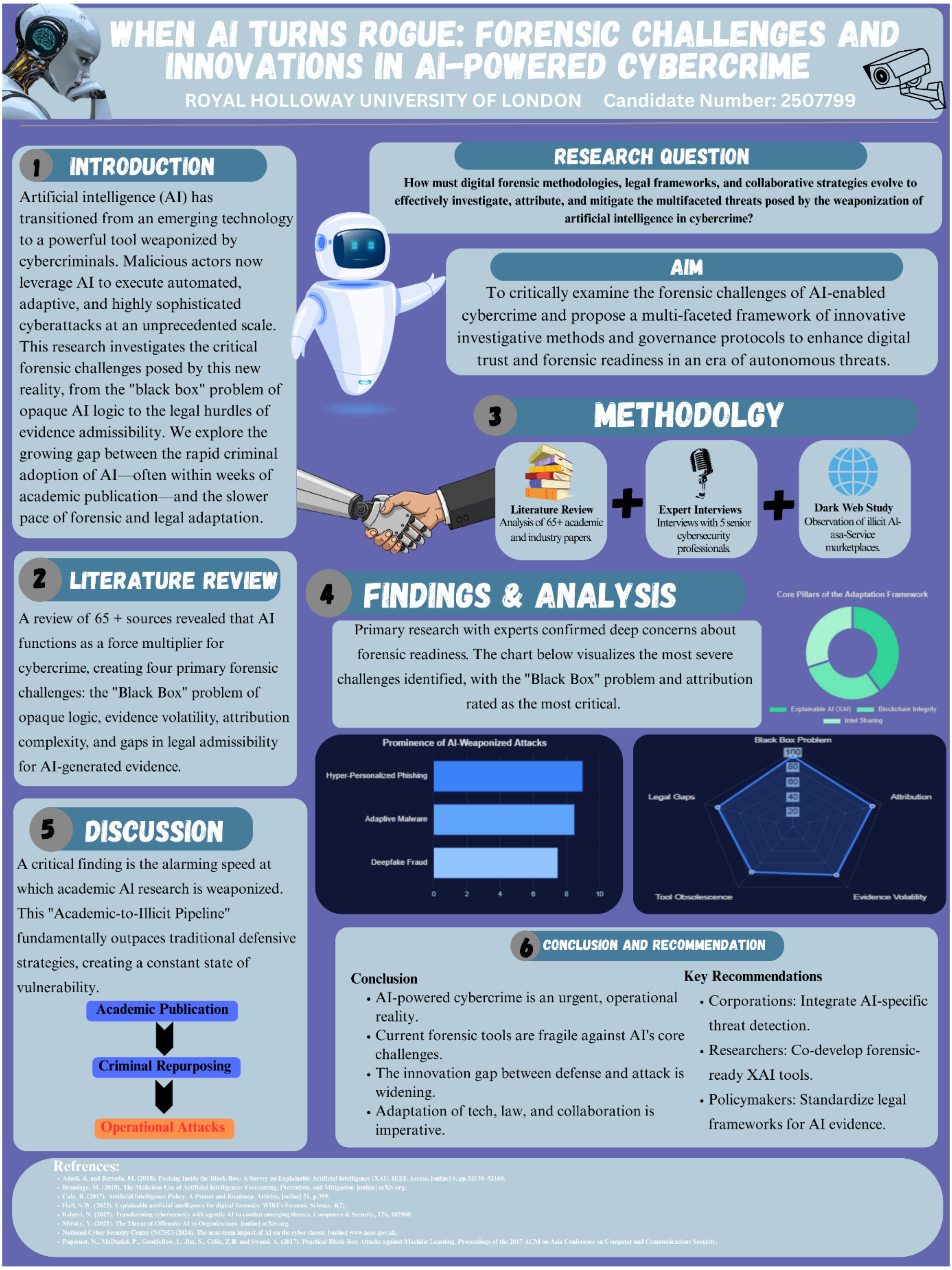
## Glossary

1. **AI (Artificial Intelligence):** Computer systems able to perform tasks normally requiring human intelligence, such as learning, problem-solving, and decision-making.
2. **Adversarial Attack:** A data point that changes the prediction of an AI model to something it should not be (usually positive or negative) when the actual data point represents the other prediction.
3. **Blockchain Provenance:** use of blockchain to capture the chain of custody and make digital forensic evidence tamper-free
4. **CaaS (Cybercrime-as-a-Service):** Commercialization (selling as a service) of cyberattacks on the dark web: AI-enabled phishing or ransomware kits for sale at a price.
5. **Dark Web:** Hidden online networks which can be accessed using a dedicated browser (such as Tor), often used for illegal activities, including the trading of AI-powered attack tools.
6. **Deepfake:** An audio, video or image produced with near or exact recognition through the use of Artificial Intelligence (AI), often using Generative Adversarial Networks (GANs) for the purpose of fraud or for spreading disinformation.
7. **Explainable AI (XAI):** AI models capable of providing transparent and understandable explanations of their predictions, important factors for future forensic and/or legal uptake of
8. **Forensic Readiness:** readiness of the organisation to gather, maintain, and present digital evidence in admissible form
9. **GANs (Generative Adversarial Networks):** AI architectures in which two neural networks (a generator and a discriminator) compete to generate believable synthetic media.
10. **Attack Tree of Threats and Techniques (MITRE ATT&CK):** A globally accepted knowledge graph that gives a mapping of malicious AI-enabled attacker tactics, techniques, and procedures (TTPs).
11. **NIST AI Risk Management Framework (NIST AI RMF):** A framework using the U.S. National Institute of Standards and Technology's Cybersecurity Framework Approach for Assessing and reducing AI Risk

12. **Polymorphic Malware:** Malware keeps evolving and mutates its code structure to deliver undetected malware, which gets more sophisticated as time goes by, with an AI assist.
13. **Rogue AI:** An AI system designed for one use may instead interact with other systems in an unintended or malignant way, creating possible obstacles to accountability in cybercrime.
14. **XAI-Forensics:** the application of AI interpretability methods (e.g. SHAP, LIME) for digital investigations to maintain the integrity and validity of the AI-generated evidence.

# Poster

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